



DPI Classification Release Notes for R8.4 and above

dpisig_2019-10-18

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Release Abstract

This Release Notes cover the applications and protocols supported by the DPI Signature in MCC R8.4 and later releases. For previous releases of MCC, refer to the *Affirmed Networks DPI Classification Release Notes for R8.3 and earlier*.

This Release Notes is a running document and is updated each time a new DPI signature file is released. The latest DPI signature file is documented first, followed in order with the previous DPI signature files. There is a base level of protocols that DPI will detect for all releases. Changes to these classifications are called out for all releases.

Upgrade Information

In general, the upgrade of a DPI signature file is hitless and does not require any reboots. At times, though, this dynamic upgrade capability cannot be supported, and the upgrade will require a reboot of SSM/WSMs for the new signature file to take effect.

If you are upgrading from an older version to a newer version, and there is a version in between that breaks the dynamic upgrade, then the SSM/WSMs will need to be rebooted for the new signature file to take effect. The DPI signature files that break dynamic upgrade and require an SSM/WSM reboot are listed below:

- dpisig_2019-10-11
- dpisig_2019-06-21
- dpisig_2019-04-05
- dpisig_2018-10-19
- dpisig_2018-06-22
- dpisig_2018-03-30
- dpisig_2018-03-09
- dpisig_2017-12-08
- dpisig_2017-09-22

Note: From MCC Release 9.4.0.0 onwards, Affirmed Networks provides a script based tool to add or update classification support for applications and protocols without any software update or a restart on MCC. For more information, refer to the *Affirmed Networks DPI Script User Guide*.

How to Obtain Product Documentation

Documentation is available for currently supported product releases. Documentation is available online in Adobe PDF format. You can view PDFs online using the Adobe Reader®. To download the latest version of the Adobe Reader software from the Adobe web site, go to <http://get.adobe.com/reader/>

Product documentation is available from the Affirmed Networks SFTP server. Contact your sales representative or the TAC for instructions on accessing the SFTP server.

See the *Affirmed Networks Documentation Library* for a complete list of Affirmed Networks product documentation.

Technical Assistance Center

Affirmed Networks Technical Assistance Center (TAC) provides technical support 24 hours a day, 7 days a week (24 x 7) for any customer with an active maintenance contract.

To contact the TAC:



E-mail: prodsupport@affirmednetworks.com



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This section describes the applications and protocols introduced, enhanced, and deprecated in the current DPI signature – dpisig_2019-10-18.

Limitations on rollback

It is not recommended to perform a DPI signature rollback to a previous version, once upgraded to the current dpisig release. This may result in an SSM crash.

If you must do a rollback/downgrade to a signature version older than dpisig_2019-08-30, follow the workaround below:

- 1 Before triggering the rollback, perform below the steps to disable EDR records generation on the setup:
 - a Note down the “**edr-interface**” parameter value in all the control-profiles.
 - b Remove “**edr-interface**” configuration from all the control-profiles.
- 2 Perform dpisig rollback and reboot all SSM/WSM/ISM nodes to bring back the previous DPI Signature in to effect.

Once all the SSM/WSM/ISM nodes are up and running after reboot, add back the **edr-interface** parameter value to all the control-profiles to re-enable EDR records generation.

Applications added

The following table lists the applications support added in the current DPI signature – dpisig_2019-10-18.

Table 1 Applications added in the current release

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Remarks
FoxNews	Fox News: Live Breaking News (iOS) Fox News: Breaking News, Live Video & News Alerts (Android)	encrypted, apache, video	Web	It is not possible to block Fox News completely. Some information, which is delivered by cloud services such as Fastly, will come through. Currently, the Video attribute is only set for HTTP traffic. Therefore, blocking of Video application attribute is not possible.
Kaltura	–	encrypted, video, audio	Advertisement and Analytic Services	Kaltura is background traffic in multiple applications.
Streamcraft	StreamCraft - Live Game Stream (iOS) StreamCraft - Live Stream Games & Chat (Android)	encrypted, amazon-cloud, video, audio	Streaming	A significant amount of traffic produced by this application (especially livestreamtraffic) will be classified as some kind of Google or YouTube traffic.

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Remarks
VerizonDigital MediaServices	–	encrypted, video	Business	Currently, the Video attribute is only set for HTTP traffic. Therefore, blocking of Video application attribute is not possible.

Protocols added

There are no new protocols added in this release.

Applications and protocols improved

The following application and protocol improvements are introduced in the current DPI signature – *dpisig_2019-10-18*:

- [Improved] *AdobeServices* application classification
- [Improved] *DisneyServices* application classification
- [Improved] *HBOServices* application classification
- [Improved] *GoogleServices* application classification
- [Improved] *LinkedIn* application classification
- [Improved] *MicrosoftServices* application classification
- [Improved] *WindowsStore* application classification
- [Improved] video application attribute for *Vudu* application classification
- [Improved] video application attribute for *Xbox* application classification
- [Improved] video application attribute for *iTunes* application classification

Applications and protocols deprecated

There are no applications or protocols deprecated in this release.

Applications and Protocols supported in the previous DPI Signature

The following tables provides the cumulative list of applications and protocols supported until the DPI signature – dpisig_2019-10-11.

Applications

Table 2 DPI Supported Applications

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
Geocaching	Geocaching (iOS) Adventure Lab (iOS) Geocaching (Android) Adventure Lab (Android)	encrypted	Gaming	Geocaching uses external services for their GPS data. For that reason, a lot of traffic caused by the app will be classified as Apple Geolocation on iOS and as Google Services on Android. Currently, the Video attribute is only set for HTTP traffic. Therefore, blocking of Video application attribute is not possible.
My5	My5 - Channel 5 (iOS) My5 (Android)	stream, web, video, apache, google, encrypted, amazon-cloud	Streaming	On iOS, parts of the My5 traffic might be classified as Akamai Cloud.
SourceEngine	–	–	Gaming	–
DisneyServices	ESPN: Live Sports & Scores (iOS) Radio Disney: Watch & Listen (iOS) ESPN (Android) Radio Disney: Watch & Listen (Android)	encrypted, apache, audio	Business	Last improved in dpisig_2019-10-18.
Level3Cloud	–	encrypted, audio, video, apache	Business	The video and audio attribute classification is based only on HTTP. Hence blocking of those attributes is not possible.
Liftoff	–	encrypted, amazon-cloud	Advertisement and Analytic Services	Liftoff is background traffic in multiple applications.
VPNSecure	VPNSecure VPN (iOS) VPNSecure (Android)	encrypted	Tunnel	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
Chartboost	–	encrypted	Advertisement and Analytic Services	Chartboost is background traffic in multiple applications.
Criteo		encrypted	Advertisement and Analytic Services	Criteo is background traffic in multiple applications.
DirecTV	DIRECTV (iOS) DIRECTV (Android)	encrypted, video, audio, apache	Streaming	Some non-streaming flows are detected as ATTServices. Help option is detected as apple-services. Currently, the attributes audio and video are only set for HTTP traffic.
Heyzap	–	encrypted	Advertisement and Analytic Services	Heyzap is background traffic in multiple applications.
Swrve	–	encrypted	Advertisement and Analytic Services	Swrve is background traffic in multiple applications.
Bebi	–	encrypted	Advertisement and Analytic Services	Bebi is background traffic in multiple applications.
AppLovin	–	encrypted	Advertisement and Analytic Services	AppLovin is background traffic in multiple applications.
DeltaDNA	–	encrypted	Advertisement and Analytic Services	DeltaDNA is background traffic in multiple applications.
AdTiming	–	encrypted	Advertisement and Analytic Services	AdTiming is background traffic in multiple applications.
Flurry	–	encrypted	Advertisement and Analytic Services	Flurry is background traffic in multiple applications.

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
Optimizely	–	encrypted	Advertisement and Analytic Services	Optimizely is background traffic in multiple applications.
SmartAdServer	–	encrypted	Advertisement and Analytic Services	SmartAdServer is background traffic in multiple applications.
Exactag	–	encrypted	Advertisement and Analytic Services	Exactag is background traffic in multiple applications.
ScorecardResearch	–	encrypted	Advertisement and Analytic Services	ScorecardResearch is background traffic in multiple applications.
Conviva	–	encrypted	Advertisement and Analytic Services	Conviva is background traffic in multiple applications.
BAMTechMedia	–	encrypted	Streaming	BAMTech Media is background traffic in multiple applications.
HBO Services	HBO GO: Stream with TV Package (iOS) HBO NOW: Stream TV & Movies (iOS) HBO Nordic (iOS) HBO GO: Stream with TV Package (Android) HBO NOW: Stream TV & Movies (Android) HBO (Android)	encrypted, stream, video, audio, apache	Streaming	The HBO Go help service uses ZenDesk. The login will be recognized as HBO Services but further traffic is classified as ZenDesk. Last improved in dpisig_2019-10-18.
HatchCloudGaming	Hatch (Android)	encrypted, amazon-cloud	Gaming	–
AdSafeProtected	–	encrypted, apache	Advertisement and Analytic Services	AdSafeProtected is background traffic in multiple applications.
AdForm	–	encrypted	Advertisement and Analytic Services	AdForm is background traffic in multiple applications.

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
FreeWheel	–	encrypted, apache, audio, video	Advertisement and Analytic Services	FreeWheel is background traffic in multiple applications. Currently, the attributes video is only set for HTTP traffic. Therefore, blocking of video application attribute is not possible.
SpotX	–	encrypted	Advertisement and Analytic Services	SpotX is background traffic in multiple applications.
IronSource	–	encrypted	Advertisement and Analytic Services	IronSource is background traffic in multiple applications.
TrafficJunky	–	encrypted, apache	Advertisement and Analytic Services	TrafficJunky is background traffic in multiple applications.
TrafficFactory	–	encrypted	Advertisement and Analytic Services	TrafficFactory is background traffic in multiple applications.
Moat	–	encrypted	Advertisement and Analytic Services	–
FaceApp	FaceApp - AI Face Editor (iOS) FaceApp - AI Face Editor (Android)	encrypted	Social	FaceApp causes a lot of advertisement-traffic, mostly from Google-Ads. So, for some scenarios, especially downloading pictures, the detection rate might not be very high.
BigFarm	Big Farm: Mobile Harvest (iOS) Big Farm: Mobile Harvest (Android)	apache, encrypted	Gaming	While playing Big Farm, the percentage of traffic classified as Big Farm will not be very high. The most traffic will be classified as Goodgame Studios, which is the publisher of Big Farm. To block Big Farm, it is therefore necessary to block also Goodgame Studios.
Asphalt8	Asphalt 8: Airborne (iOS) Asphalt 8: Airborne (Android)	encrypted	Gaming	When launching this app background traffic [vungle,unity,stackpath-cloud] and facebook-cloud flows are seen. These are not correlated with the app. For correlation issue we are tracking in AN-46767.

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
SOMA	SOMA Messenger (iOS) SOMA free video call and chat (Android)	encrypted	Messaging	The SOMA application produces high amounts of advertisement and background traffic, mostly by Google-Ads and Unity. Therefore, the amount of traffic that is classified as SOMA can be quite low in comparison.
Adcolony	–	encrypted	Advertisement and Analytic Services	Since Adcolony is an analytics platform, it will be part of many different services as background traffic. Last improved in dpisig_2019-10-11.
Mixpanel	–	encrypted	Advertisement and Analytic Services	Mixpanel is background traffic in multiple applications, like Viber. Last improved in dpisig_2019-10-11.
InventiveVPN Services	VPN For iPhone (iOS) VPN - Best VPN Unlimited Proxy (iOS)	apache, encrypted	Tunnel	Starting the applications will not be classified as Inventive VPN Services. Last improved in dpisig_2019-10-11.
Mudfish	Mudfish Cloud VPN (Android)	encrypted	Tunnel	SSL Session ID Tracking (SIT) is required for the detection to work properly. If a game is successfully connected via Mudfish and a multiplayer game is started, some generated traffic may not use the Mudfish protocol.
LordsMobile	Lords Mobile: War Kingdom (iOS) Lords Mobile: War Kingdom (Android)	apache, encrypted	Gaming	Linking FB account/google account are not correlated as LordsMobile. The game provides access to fb/twitter/youtube pages to see the game related news and posts. These flows are also not correlated to LordsMobile. Parts of the traffic is classified as IGG, because it is the publisher of Lords Mobile. Traffic correlation issues are being tracked with AN-46767.

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
IGG	Lords Mobile: War Kingdom (iOS) Mobile Royale: Kingdom Defense (iOS) Castle Clash: War Empire (iOS) Clash of Lords 2: Guild Castle (iOS) Conquerors: Golden Age (iOS) Lords Mobile: War Kingdom (Android) Mobile Royale MMORPG (Android) Castle Clash: Brave Squads (Android) Clash of Lords 2: Guild Castle (Android) Conquerors: Golden Age (Android)	apache, encrypted	Gaming	Some of the games of IGG use Google or Facebook for certain functionalities. Also, a lot of advertisements will lower the detection rate. Last improved in dpisig_2019-10-11.
Tapjoy	–	encrypted, apache, video	Advertisement and Analytic Services	Tapjoy will be classified mostly as part of background traffic of other applications. It has no own application. Last improved in dpisig_2019-10-11.
DeadRivals	Dead Rivals - Zombie MMO (iOS) Dead Rivals - Zombie MMO (Android)	encrypted	Gaming	The correlation of unity and game-kit flows need to be added. Parts of the traffic will be classified as Gameloft Services. Traffic correlation issues are being tracked with AN-46767.
SnapMasterVPN	Free VPN & security unblock Proxy -Snap Master VPN (Android) VPN Proxy Master lite - free&secure VPN proxy (Android) Free unblock VPN& security VPN (Android) VPN Proxy Master (iOS)	encrypted, amazon-cloud	Tunnel	Last improved in dpisig_2019-10-11.
Joyn	Joyn deine Streaming App (iOS) Joyn deine Streaming App (Android)	encrypted, apache	Streaming	Parts of the Joyn traffic may be classified as Pro7Sat1 Services.
Pro7Sat1Services	Joyn deine Streaming App (iOS) Joyn deine Streaming App (Android)	encrypted, apache	Streaming	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
UnblockSites	Unblock Sites (Android)	apache	Tunnel	–
NexxTV	nexxCANVAS (Android)	encrypted	Business	–
Funk	funk (iOS) funk – Webserien, Dokus, Vlogs, Shows (Android)	encrypted	Streaming	Nexx-tv flows are generated by this App. In iOS, the funk trailer option generates Youtube and Instagram flows. These flows are not correlated to funk. Traffic correlation issues are being tracked with AN-46767.
CounterpathServices	–	encrypted	Voice over IP	–
ATTServices	DIRECTV NOW (iOS) AT&T WatchTV (iOS) U-verse (iOS) DIRECTV NOW (Android) AT&T WatchTV (Android) AT&T U-verse (Android)	encrypted, video, audio, apache	Streaming	Currently, the attributes audio and video are only set for HTTP traffic. Last improved in dpisig_2019-10-11.
BrawlStars	Brawl Stars (iOS) Brawl Stars (Android)	encrypted	Gaming	To block Brawl Stars, it is necessary to also block Supercell, because in blocking scenarios the traffic between both is not distinguishable anymore
WizardsUnite	Harry Potter: Wizards Unite (Android) Harry Potter: Wizards Unite (iOS)	encrypted, apache	Gaming	Parts of the traffic will be classified as Niantic Labs, because this is the publisher of Wizards Unite.
AdobeServices	Adobe Creative Cloud (Android) Adobe Connect (Android) Adobe Creative Cloud (iOS) Adobe Connect (iOS)	encrypted, apache, file-transfer, video	Business	Last improved in dpisig_2019-10-18.
Zoom	ZOOM Cloud Meetings (Android) ZOOM Cloud Meetings (iOS)	encrypted	Conference	–
Radio-De	radio.net (Android) radio.net (iOS)	encrypted, audio, apache	Streaming	Feedback and FAQs are detected as Zendesk. These flows are not correlated to radio-de. Some of the traffic is detected as Facebook-cloud, which is not correlated to the application. Traffic correlation issues are being tracked with AN-46767.
UnrealEngine	Fortnite (iOS)		Gaming	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
BBCNews	BBC News (Android) BBC News (iOS)	encrypted, apache, video	Streaming	When connected through outlook or gmail, the traffic is detected as outlook / gmail. At times, some traffic is detected as Adjust, iplayer, and yandex packets. When BBC news application is opened, BBC news, BBC services, Google ads, and google analytics are detected. When we check other BBC options, google play and google services are detected. Help is detected as google apis. These flows are not correlated. Traffic correlation issues are being tracked with AN-46767. Last improved in dpisig_2019-07-19.
BBCServices	Forge of Empires (Android)	encrypted, apache, audio, video, google	Business	Last improved in dpisig_2019-10-11.
InnoGames	Elvenar (Android) God Kings (Android) Warlords of Aeternum (Android) Grepolis - Divine Strategy MMO (Android) Tribal Wars (Android) Tribal Wars 2 (Android) Forge of Empires: Build a City (iOS) Elvenar - Fantasy Kingdom (iOS) God Kings (iOS) Warlords of Aeternum (iOS) Grepolis - Divine Strategy MMO (iOS) Tribal Wars (iOS) Tribal Wars 2 (iOS)	encrypted	Gaming	Last improved in dpisig_2019-07-19
ForgeofEmpires	Forge of Empires (Android) Forge of Empires: Build a City (iOS)	encrypted, audio	Gaming	Currently, the attribute audio is only set for HTTP traffic. Therefore, blocking of the audio application attribute is not possible.
SpiegelTV	SPIEGEL.TV (Android) SPIEGEL.TV (iOS)	encrypted, video, apache	Streaming	For Android some traffic will be classified as Spiegel Services or Nexx Cloud. For iOS and Brwoser almost all traffic will be classified as Spiegel Services or Nexx Cloud. These flows are not correlated. Last improved in dpisig_2019-08-02.

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
SpiegelServices	–	encrypted, video, apache	Web	–
ReceiverRadio	Receiver Radio (Android) Receiver Radio (iOS)	encrypted, audio, video, apache	Streaming	The correlation of the background flows when loading the app and browsing the app with the application flows is not yet supported. There is an option to view home page of the radio stations, but these flows are not correlated to the application. Traffic correlation issues are being tracked with AN-46767. Last improved in dpisig_2019-07-19
EurosportPlayer	Eurosport Player - Live Player Streaming (Android) Eurosport Player (iOS)	encrypted, audio, video	Streaming	The attributes audio and video are only set for HTTP traffic. Some flows of the help option is detected as unknown and some are detected as Zendesk. These flows are not correlated to the application traffic. Traffic correlation issues are being tracked with AN-46767. Last improved in dpisig_2019-10-11.
GoodgameStudios	Big Farm: Mobile Harvest - Free Farming Game (Android) Empire: Four Kingdoms (Android) Skytopia - City Tycoon (Android) Empire: Millennium Wars (Android) Big Farm: Mobile Harvest (iOS) Empire Four Kingdoms (iOS) BigCompany: Skytopia (iOS) Empire: Millennium Wars (iOS)	encrypted, apache	Gaming	For some games developed by Goodgame Studios, the classification rate will be quite low because they use the Unity engine. Therefore most traffic will be classified as Unity.
ElectronicArts	Electronic Arts	–	Gaming	This is an umbrella classification to capture traffic belonging to this publisher. It includes traffic for related websites and background traffic from games/services, that are not separable into dedicated applications.
Radioplayer	Radioplayer - Radio für alle (iOS) Radioplayer - Gratis Radio App (Android)	encrypted	Streaming	There is a buy option, which is detected as amazon. These flows are not correlated to application flows. When we try to access some stations, it displays '404 not found' error. Traffic correlation issues are being tracked with AN-46767.

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
MoPub	MoPub	encrypted	Advertisement and Analytic Services	MoPub is part of many applications as background traffic Last improved in dpisig_2019-10-11.
IngressPrime	Ingress Prime (iOS) Ingress Prime (Android)	encrypted, google	Gaming	There is a lot of additional traffic classified as Google Cloud, because of google service usage
PUBGMobile	PUBG MOBILE (iOS) PUBG MOBILE (Android)	encrypted	Gaming	Parts of the traffic will be classified as Tencent Games, because Tencent is the developer and publisher of PUBG Mobile.
7TV	Joyn deine Streaming App (iOS) Joyn deine Streaming App (Android)	encrypted, apache	Deprecated	–
IPVanish	IPVanish VPN: The Fastest VPN (iOS) IPVanish VPN (Android)	encrypted	Tunnel	–
TunnelBear	TunnelBear VPN & Wifi Proxy (iOS) TunnelBear VPN (Android)	encrypted, amazon-cloud	Tunnel	–
ZeroVPN	Zero VPN (Android)	encrypted	Tunnel	–
ZenmateVPN	ZenMate VPN und Proxy-Server (iOS) ZenMate VPN - WiFi VPN Security & Unblock (Android)	encrypted	Tunnel	It is not possible to successfully block Zenmate VPN.
TigerVPN	tigerVPN - Fast & Secure VPN (iOS) VPN by tigerVPN (Android)	encrypted	Tunnel	–
Surfeasy	SurfEasy VPN (iOS) SurfEasy Secure Android VPN (Android)	encrypted	Tunnel	–
PrivateInternetAccess	VPN - Private Internet Access (iOS) VPN by Private Internet Access (Android)	encrypted	Tunnel	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
VPNMaster	VPN Master - Free VPN & security unblock Proxy (Android)	encrypted	Deprecated	–
SnapVPN	Snap VPN - Unlimited Free & Super Fast VPN Proxy (Android)	encrypted	Tunnel	–
TurboVPN	Turbo VPN Private Browser (iOS) Turbo VPN – Free Unlimited VPN & Secure Hotspot (Android)	encrypted	Tunnel	–
AviraPhantomVPN	Avira Phantom VPN & WiFi Proxy (iOS) Avira Phantom VPN: Free & Fast VPN Client & Proxy (Android)	encrypted	Tunnel	On Android, the detection will only work if the VPN connection is started a few seconds after opening the application. Blocking works for all scenarios.
HMAVPN	HMA! Hotspot VPN & Proxy (iOS) HMA! VPN Proxy & WiFi Security, Online Privacy PRO (Android)	encrypted	Tunnel	–
VPNUnlimited	VPN Unlimited for iPhone, iPad (iOS) KeepSolid VPN Unlimited Free VPN for Android (Android)	encrypted	Tunnel	–
NordVPN	NordVPN: VPN Fast & Unlimited (iOS) VPN: Fast, Secure & Unlimited NordVPNd (Android)	encrypted, google	Tunnel	–
CyberGhost	CyberGhost VPN & WiFi Proxy (iOS) CyberGhost VPN - Fast & Secure WiFi protection (Android)	encrypted	Tunnel	–
ExpressVPN	ExpressVPN - #1 Trusted VPN (iOS) ExpressVPN - Best Android VPN (Android)	encrypted	Tunnel	If the user switches between servers after establishing a connection, all re-establishing connections beyond the first server cannot be classified. Last improved in dpisig_2019-08-30.
HideMeVPN	hide.me VPN (iOS) hide.me VPN (Android) HideMe VPN - Inceptial (Android)	encrypted	Tunnel	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
ProtonVPN	ProtonVPN - Fast & Secure VPN (iOS) ProtonVPN – advanced online security for everyone (Android)	encrypted	Tunnel	–
TunnelGuru	Troid VPN Free VPN Proxy (Android) Mini WebTunnel (Android)	encrypted, apache	Tunnel	–
SaferVPN	SaferVPN - Best VPN for WiFi (iOS) Free VPN Proxy & WiFi security - SaferVPN (Android)	encrypted	Tunnel	–
FrootVPN	Froot VPN (Android, iOS)	encrypted	Tunnel	Blocking of FrootVPN is only supported for the OpenVPN tunnel modes (both tcp and udp). The IPSec tunnel mode traffic can be classified but not blocked.
WindscribeVPN	Windscribe VPN (iOS) Windscribe VPN (Android)	encrypted	Tunnel	–
IVPN	IVPN - Privacy & Security VPN (iOS) IVPN (Android)	encrypted	Tunnel	–
HiVPN	VPN – Best Unlimited VPN Proxy (iOS) Hi VPN, Free VPN – Fast, Secure and Unlimited VPN (Android)	encrypted	Tunnel	–
PureVPN	VPN for Iphone By PureVPN (iOS) PureVPN - Secure & Best Free VPN for Android (Android)	encrypted	Tunnel	–
VPNProxyMaster	VPN Proxy Master (iOS) VPN Proxy Master (Android)	encrypted	Deprecated	–
FireVPN	Free VPN by FireVPN (Android)	encrypted	Tunnel	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
StrongVPN	StrongVPN — The Strongest VPN (iOS) StrongVPN - Unlimited Privacy (Android)	encrypted	Tunnel	–
MullvadVPN	Mullvad VPN (Android & iOS) MullvadVPN- OpenVPN for Android (Android)	encrypted	Tunnel	Only the OpenVPN-mode of Mullvad VPN is covered by the classification. The Wireguard mode will be covered by an own protocol in the future
FreedomVPN	F-Secure FREEDOME VPN (iOS) FREEDOME VPN Unlimited anonymous Wifi Security (Android)	encrypted	Tunnel	–
IvacyVPN	Ivacy VPN (iOS) Ivacy VPN - Best Fast VPN (Android)	encrypted	Tunnel	–
ZOOGVPN	ZoogVPN - Unblock & Privacy (iOS) Zoog VPN - Internet freedom, security and privacy (Android)	encrypted	Tunnel	–
BufferedVPN	Buffered VPN (iOS) Buffered VPN (Android)	encrypted	Tunnel	–
VPNArea	VPNArea (iOS) VPN Area: Best VPN for Android (Android)	encrypted, apache	Tunnel	–
SpeedVPNInventive	VPN - Best VPN Unlimited Proxy (iOS)	encrypted, apache	Deprecated	–
SuperVPN	SuperVPN Free VPN Client (Android)	encrypted	Tunnel	–
SteganosOnlineShieldVPN	mySteganos Online Shield VPN (iOS) mySteganos Online Shield VPN (Android)	encrypted	Tunnel	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
HolaVPN	Hola Privacy VPN App (iOS) Hola Free VPN Proxy (Android)	encrypted, google	Tunnel	Hola VPN will not be detected on Android because it uses an HTTP proxy. We don not cover classification of proxy servers. Instead, we detect the applications which shall be hide by the proxy. Blocking on android is not supported for this app as classification is having limitation
ibVPN	123VPN - Simple VPN (iOS) 123VPN - FREE VPN (Android) ibVPN - VPN for Wifi Security (iOS) ibVPN - Unlimited VPN app (Android)	encrypted	Tunnel	–
SpeedVPNMobileAppDev	Speed VPN by MobileAppDev (Android)	encrypted, apache	Tunnel	–
ShellfireVPN	Shellfire VPN (iOS) Shellfire VPN (Android)	encrypted	Tunnel	–
AtomVPN	Atom VPN (Android)	encrypted, apache	Tunnel	–
.ArkVPN	ArkVPN: Standard Edition (iOS)	encrypted, apache	Tunnel	–
AmazeVPN	Amaze VPN (Android)	encrypted	Tunnel	This app is getting blocked partially. This issue is being tracked.
FinchVPN	Free & Premium VPN - FinchVPN (Android)	encrypted	Tunnel	–
FastSecureVPN	Fast Secure VPN (Android)	encrypted	Tunnel	–
VPNEasy	VPN Easy (Android)	encrypted	Tunnel	–
.ZPNVPN	.Free VPN Proxy - ZPN (Android)	encrypted	Tunnel	–
AppVPN	VPN Unlimited Proxy AppVPN (iOS) VPN Unlimited Proxy AppVPN (Android)	encrypted	Tunnel	The IPSec-Mode of AppVPN is currently not supported

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
X-VPN	X-VPN Unlimited VPN Proxy (iOS) X-VPN - Free Unlimited VPN Proxy (Android)	encrypted, google	Tunnel	This app is not getting blocked. This issue is being tracked.
DotVPN	DotVPN — a better way to VPN (iOS) DotVPN — better than VPN (Android)	encrypted	Tunnel	—
LinkVPN	LinkVPN - Fast & Unlimited VPN Proxy (iOS) LinkVPN Free VPN Proxy (Android)	encrypted	Tunnel	—
DroidVPN	DroidVPN - Easy Android VPN (Android)	encrypted	Tunnel	—
BrowsecVPN	Browsec VPN - Unlimited VPN (iOS) Browsec VPN - Free and Unlimited VPN (Android)	encrypted	Tunnel	—
Encrypt-me	Encrypt.me (iOS) Encrypt.me - Super Simple VPN (Android)	encrypted	Tunnel	—
SuperVPNbyCheng	SuperVPN - Best Free Privacy & Security Proxy (iOS)	encrypted	Tunnel	—
VPN365	VPN 365 - Ip Changer & Unblock (iOS)	encrypted	Tunnel	—
HotspotVPN	VPN - Master Proxy for iPhone (iOS) VPN - Proxy Master Unlimited (iOS) Hotspot VPN - Super Free VPN Unlimited Proxy (Android) Free VPN Unlimited Proxy - Proxy Master (Android) Secure VPN - Free VPN Proxy, Best & Fast Shield (Android) Easy VPN - Free VPN proxy master, super VPN shield (Android)	encrypted	Tunnel	—

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
TikTok	TikTok - Real Short Videos (iOS) TikTok (Android)	video, encrypted, audio, amazon-cloud	Social	Last improved in dpisig_2019-10-11.
Megogo	Megogo.net (iOS) MEGOGO –ТВ и Кино (Android)	encrypted, video	Streaming	–
Fortnite	Fortnite (iOS)	encrypted	Gaming	Last improved in dpisig_2019-08-30.
YandexMaps	Yandex.Maps (iOS) Yandex.Maps (Android)	encrypted	Business	Some of the Yandex Maps traffic is detected as Yandex with the attribute encrypted.
YandexMail	Yandex.Mail - Email App (iOS) Yandex Mail (Android)	encrypted	Mail	Some of the <i>Yandex Mail</i> traffic is detected as <i>Yandex</i> with the attribute <i>encrypted</i> .
YandexDisk	Yandex.Disk saves your photos (iOS) Yandex Disk (Android)	encrypted	FileTransfer	Some of the <i>Yandex Disk</i> traffic is detected as <i>Yandex</i> with the attribute <i>encrypted</i> .
YandexNavigator	Yandex.Navi (iOS) Yandex.Navigator (Android)	encrypted	Business	Some traffic from <i>Yandex Navigator</i> will be classified as Yandex Maps, due to it's use of related services
YandexMusic	Yandex Music (iOS) Yandex Music (Android)	encrypted	Streaming	Some of the <i>Yandex Music</i> traffic is detected as <i>Yandex</i> with the attribute <i>encrypted</i> .
Discord	Discord (iOS) Discord (Android)	encrypted, audio	VoiceoverIP	–
YoutubeMusic	YouTube Music (iOS) YouTube Music (Android)	encrypted, audio, stream	Streaming	Youtube Music is based on Youtube. Therefore, the login is classified as Youtube. There is no separation between music and music video flows possible. All related traffic will be classified by the audio attribute. Blocking of Youtube Music is not possible. The application will fall back to generic Youtube behavior.

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
WireMessenger	Wire • Secure Messenger (iOS) Wire • Secure Messenger (Android)	video, encrypted, audio	Messaging	In a block test scenario, the GUI gives you the impression of an established audio connection (connection timer increases, ...) even if the audio data is blocked.
Maxdome	maxdome (iOS) maxdome (Android)	encrypted, video, stream, audio, apache	Streaming	The audio attribute is set for unencrypted HTTP traffic. Currently Maxdome only uses encrypted traffic.
AmazonDrive	Amazon Drive (iOS) Amazon Drive (Android)	encrypted, amazon-cloud	Business	–
FoxSports	FOX Sports (iOS) FOX Sports (Android)	encrypted, apache, amazon-cloud, video, google	Streaming	Last improved in dpisig_2019-10-11.
Houseparty	Houseparty (iOS) Houseparty (Android)	encrypted, video, audio	Social	only a fraction of the traffic gets classified since houseparty uses tokbox, OpenTok for voice audio/video transfers.
GoogleCalendar	Google Calendar (iOS) Google Calendar (Android)	encrypted, google	Business	Most of the <i>Google Calendar</i> traffic is detected as <i>Google Shared Services</i> with the attribute <i>encrypted</i> . It is not possible to block <i>Google Calendar</i> .
SlingTV	Sling TV (iOS) Sling TV (Android)	encrypted, stream, apache	Streaming	Last improved in dpisig_2019-10-11.
Wish	Wish (iOS) Wish (Android)	encrypted, video	Business	–
Vudu	Vudu - Movies & TV (iOS) Vudu - Rent, Buy or Watch...(Android)	encrypted, apache, video, audio	Web	Last improved in dpisig_2019-10-18.
Sarahah	Sarahah (Android)	encrypted	Social	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
GoogleDuo	Google Duo (iOS) Google Duo (Android)	apache	VoiceoverIP	Blocking of Google Duo is not possible because in blocking scenarios, Google Duo uses generic Google patterns.
Viber	Viber (iOS) Viber Messenger (Android)	file-transfer, encrypted, audio, chat, video, amazon-cloud	VoiceoverIP	If only the protocol <i>Viber</i> is blocked, the application <i>Viber</i> starts an SSL connection and is therefore still running. It can be blocked by blocking the application <i>Viber</i> . Limitations for blocking the application <i>Viber</i> with specific application attributes apply for the protocol as well, as the application uses the protocol <i>Viber</i> . Last improved in dpisig_2019-08-30.
Facebook	Facebook (iOS) Facebook (Android) Messenger (iOS) Messenger (Android) Messenger Lite (iOS) Messenger Lite (Android)	apache, video, encrypted, chat, file-transfer, web, stream, audio, google, amazon-cloud	Social	Blocking incoming messages is not always possible. For Browser, the transfer of pictures cannot be blocked because there are generic patterns used for this. Live video can only be detected as video in the outgoing direction. Incoming live video traffic is covered by the Facebook cloud application. Embedded video content may be classified as filetransfer. On iOS or Android, it is not possible to differentiate between generic Facebook and chat traffic. Therefore, chat traffic on iOS or Android will be classified as Facebook encrypted. Last improved in dpisig_2019-08-02.
SkypeforBusiness	Skype for Business (iOS) Skype for Business (Android)	encrypted, file-transfer, audio, video, google	Business	Detection of switching back from Video to voice call is not supported.

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
WhatsApp	WhatsApp (iOS) WhatsApp (Android)	audio, encrypted, file-transfer, google, video, chat	Messaging	Apple mobile devices are capable of receiving WhatsApp messages even when the App is not started. This is not WhatsApp specific traffic. Apple uses iMessage to relay the messages. Inside a Browser it is not possible to block file-transfers which are smaller than 200 KBytes. In blocking scenario of file-transfer, after sending or receiving a file, also chat messages will be blocked for the current conversation. To block file-transfers properly, it is necessary to close WhatsApp via the application manager, not via a swipe movement.
NetFlix	Netflix (iOS) Netflix (Android)	video, stream, apache, encrypted, audio	Streaming	Only streaming content and the website after login will be detected as Netflix. HTTP traffic, which contains preview images, will not be detected as Netflix. Because of this, browsing the website could be possible in a block scenario, whereas streaming content is still blocked.

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
Skype	Skype for iPhone (iOS) Skype (Android)	encrypted, audio, out, video, file-transfer	VoiceoverIP	Skype connections in general are not exclusively used for a single type of connections. That means that video and voice can be transferred in the same or separate flows. So if a video flow is detected, the same flow may also contain voice packets. Also, file transfers can be using separate flows or can be multiplexed in an existing voice call. The Skype client is also capable of instant messaging with a Facebook account. This chat can't get safely detected as Skype without causing false positives, because it uses the Facebook API. PACE classifies the traffic as protocol XMPP with Application Facebook. Voice and video calls are classified after some initial time and the type will be reevaluated recurrently. That means, that a flow marked as sub protocol unknown can be marked as voice later and switch again if video is detected because it is possible to enable video transmission at any given time during a phone call. Traffic to gateway.messenger.live.com is detected as Skype due to the integration of MSN into the Skype client. Skype and Skype for Business share the same technology and as such their traffic cannot always be distinguished. When possible both protocols should be treated as one to avoid issues arising from this. Last improved in dpisig_2019-10-11.

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
YouTube	Youtube (iOS) Youtube (Android)	audio, apache, video, stream, encrypted, google	Streaming	Every access to the <i>Youtube</i> website via <i>HTTP</i> or <i>HTTPS</i> will be detected as application <i>Youtube</i> , this also includes applications using the <i>Youtube</i> API. In some cases, an access on Android devices to the website via <i>HTTPS</i> cannot be detected as application <i>Youtube</i> , because the exchanged certificate contains a generic Google name. Streaming content is still detected as application <i>Youtube</i> in this case. Streaming video content from the domain <i>googlevideo.com</i> is also classified as application <i>Youtube</i> . In cases where a differentiation of web and streaming traffic is required this can be achieved by checking the application ID in combination with the real protocol and the real sub-protocol. If application <i>Youtube</i> was detected and the real protocol is <i>HTTP</i> with the real sub-protocol being either <i>HTTP media</i> , <i>HTTP audio</i> or <i>HTTP video</i> it is streaming traffic. The values for application ID, real protocol and real sub-protocol can be obtained by calling the public function <i>ipoque_pace_get_detection_result</i> .
QQ	QQ (iOS) QQ (Android)	encrypted, audio, video, file-transfer, apache, google	Messaging	It is possible that WeChat is classified, because WeChat and QQ belong to the same company called Tencent. Both using same pattern and it is hard to distinguish between both. Last improved in dpisig_2019-08-30.
Tango	Tango (iOS) Tango (Android)	encrypted, audio, video, file-transfer, chat, apache, amazon-cloud, stream	VoiceoverIP	Last improved in dpisig_2019-10-11.
Twitter	Twitter (iOS) Twitter (Android)	apache, web, video, encrypted, audio, google	Social	—

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
Line	Line (iOS) Line (Android)	audio, video, encrypted, apache, google	VoiceoverIP	–
Instagram	Instagram (iOS) Instagram (Android)	video, encrypted, file-transfer, chat, amazon-cloud, apache	Social	Chat cannot be separated entirely from other traffic because of the usage of generic services. In rare cases the detection rate is small in comparison to the loading of videos and pictures when opening the application. Viewing pictures may be classified as generic, because this is the main purpose of <i>Instagram</i>, but also as encrypted file transfer. Live video can only be detected as video in the outgoing direction. Incoming live video is classified as encrypted file transfer. Also, it is possible that live videos are transferred over generic Facebook flows. These flows will be classified as Facebook. Last improved in dpisig_2019-10-11.
Spotify	Spotify (iOS) Spotify (Android)	encrypted, stream, amazon-cloud, audio, video, apache	Streaming	–
2Gis	2Gis (iOS) 2GIS (Android)	encrypted	Business	This application produces a lot of additional advertisement traffic; therefore the classification rate might be low.
WeChat	WeChat (iOS) WeChat (Android)	chat, file-transfer, web, encrypted, video, audio, apache, google	Messaging	If the audio attribute is blocked, no audio or video calls can be made or received. Blocking the chat attribute will block the whole app. Last improved in dpisig_2019-10-11.
1kxun	1kxun (iOS) Q Video (Android)	apache, encrypted	Streaming	–
Alibaba	Alibaba (iOS) Alibaba (Android)	encrypted, apache, audio, video	Business	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
AliPay	AliPay (iOS) AliPay (Android)	encrypted, apache, video	Business	–
AirBnB	AirBnB (iOS) AirBnB (Android)	encrypted	Business	–
AmazonChime	Amazon Chime (iOS) Amazon Chime (Android)	amazon-cloud, video, encrypted, audio, chat, file-transfer	Business	The chat is relying on traffic used for file-transfer and thus is blocked with it. The audio-traffic is detected, but on platforms other than Android and iOS it cannot always be blocked
BOTIM	BOTIM - video calls and chat (iOS) BOTIM - Unblocked Video Call and Voice Call (Android)	encrypted	VoiceoverIP	BOTIM uses Facebook to send parts of its traffic.
BlackBerryMessenger	BBM (iOS) BBM (Android)	encrypted, apache	VoiceoverIP	<i>BlackBerry Messenger</i> sends most parts of its traffic via <i>Google Cloud</i> for <i>file-transfer</i> .
CME	C'Me - voice and video Calls (iOS) C'Me - Voice & Video Calls (Android)	encrypted	VoiceoverIP	–
YeeCall	YeeCall (iOS) YeeCall (Android)	encrypted, amazon-cloud	VoiceoverIP	Last improved in dpisig_2019-08-30.
Gaana	Gaana Music (iOS) Gaana Music (Android)	audio, video, amazon-cloud, encrypted	Streaming	<i>Gaana</i> uses <i>Akamai Cloud</i> as content provider. Therefore streaming and download content will be classified as <i>Akamai Cloud</i> .
EpicGames	Epic Browser	encrypted	Gaming	Last improved in dpisig_2019-08-30.
TVPlayer	TVPlayer TVPlayer (Android)	encrypted, video	Streaming	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
TokBox	OpenTokRTC (iOS) OpenTokApp (iOS) OpenTokApp (Android)	encrypted	Messaging	–
AfterSchool	After School (iOS)	amazon-cloud, video, encrypted, apache	Deprecated	–
AliExpress	AliExpress Shopping App (iOS) AliExpress (Android)	encrypted, apache	Business	–
Amazonshopping	Amazon (iOS) Amazon Shopping (Android)	encrypted, audio, apache, google, stream	Business	Some of the Amazon shop traffic may be detected as Amazon Cloud instead, if it is not distinguishable. This affects SSL flows.
AgeofWarringEmpire	Age of Warring Empire (iOS) Age of Warring Empire (Android)	–	Streaming	This application produces a lot of additional advertisement traffic. Therefore, the basic detection rate is low in comparison to the caused traffic.
AmazonPrimeMusic	Amazon Music (iOS) Amazon Music (Android)	Chat, file-transfer, web, encrypted, video, audio, apache	Streaming	Some of the traffic will be detected as Amazon Cloud. For that reason, blocking Amazon Prime Music completely on Mac OSX or Windows is not possible.
Anghami	Anghami (iOS) Anghami (Android)	encrypted, audio, video, amazon-cloud, apache	Streaming	–
AnyDesk	AnyDesk (iOS) AnyDesk (Android)	encrypted	RemoteControl	–
AmazonPrimeVideo	Amazon-Prime-Video (iOS) Amazon-Prime-Video (Android)	audio, encrypted, stream, amazon-cloud	Streaming	Some of the traffic will be detected as Amazon Cloud. For that reason, blocking Amazon Prime Music completely on Mac OSX or Windows is not possible.

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
AngryBirds	AB Classic (iOS) angry Birds Classic (Android)	apache, google, video, amazon-cloud	Gaming	This application produces a lot of additional advertisement traffic. Therefore, the basic detection rate is low in comparison to the caused traffic.
Azar	Azar (iOS) Azar (Android)	encrypted, apache, google, amazon-cloud	Social	Azar works in cooperation with Facebook. Therefore, a high amount of UDPtraffic while video-chatting within Azar is classified as Facebook.
AviraAntivirus	Avira Mobile Security (iOS) Avira Antivirus Security 2019 (Android)	apache	Business	–
BackBlaze	BackBlaze (iOS) BackBlaze (Android)	encrypted	Business	–
Badoo	Badoo (iOS) Badoo (Android)	encrypted	Social	–
Auth0	Auth0 Guardian (iOS) Auth0 Guardian (Android)	encrypted	Business	–
Audiko	Audiko (iOS) Audiko (Android)	audio,video	Streaming	–
BattleNet	Blizzard battle.net (iOS) Blizzard battle.net (Android)	encrypted, apache, google, audio, video	Gaming	–
ICQ	ICQ – Video Call, Text & Voice (iOS) ICQ – Video Calls & Chat App (Android) ICQ - Video Call & Chat Rooms (Android) ICQ: Messages, Group chats & Video Calls	apache, file-transfer, audio, encrypted, video, google	Messaging	ICQ is owned by Mail-Ru. Therefore, some generic traffic will be classified as Mail-Ru.

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
4shared-com	4shared Mobile (iOS) 4shared (Android)	ddl, apache, encrypted, video	Sharehosting	–
VeePN	Unlimited VPN Proxy by VeePN (iOS) Fast VPN proxy by VeePN (Android)	encrypted	Tunnel	–
JustProxyVPN	Just Proxy VPN (Android)	encrypted	Tunnel	–
360MobileSecurity	360 Security (Android)	encrypted, apache	Business	For Android and Mac OSX, a blocking of the main application functionality is not possible, because the network traffic of 360 total security only consist of signature updates and login mechanisms. For the same reason the actual detected traffic will not be high
AdobeConnect	Adobe Connect (iOS) Adobe Connect (Android)	–	Business	–
ActVPN	Act VPN (Android)	encrypted	Tunnel	–
AdobeCreativeCloud	Adobe Creative Cloud (iOS) Adobe Creative Cloud (Android)	encrypted, apache	Web	The login should not be blocked, because it is a generic login for different adobe services. Introductory tutorial videos belong to Adobe TV and will therefore not be classified as Adobe Creative Cloud. Last improved in dpisig_2019-10-11.
Adrive	ADriveMobile (Android)	audio, encrypted, apache, google, video	File Transfer	–
AppleMusic	AppleMusic (iOS)	audio, video, encrypted, stream	Streaming	Most of the Apple Music traffic is detected as iTunes or Apple services. This traffic is higher on mobile devices. In these cases it is not possible to block Apple Music. The video attribute classification for Apple Music is based on HTTP and does not work for TLS traffic. Last improved in dpisig_2019-10-11.

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
AVGAntivirus	AVG AntiVirus 2019 for Android Security FREE (Android)	apache	Business	–
Booking-com	Booking.com Travel Deals (iOS) Booking.com Travel Deals (Android)	encrypted, apache	Web	–
CandyCrush	Candy Crush Saga (iOS) Candy Crush Saga (Android)	encrypted, amazon-cloud	Gaming	–
CandyCrushSoda	Candy crush soda (iOS) Candy crush soda (Android)	encrypted	Gaming	–
BubbleWitch	Bubble Witch Saga (iOS) Bubble Witch Saga (Android)	encrypted	Gaming	–
Box	Box for iphone and ipad (iOS) Box (Android)	encrypted	File Transfer	–
BoomBeach	Boom Beach (iOS) Boom Beach (Android)	encrypted	Gaming	Boom Beach generates a significant amount of traffic by the Supercell protocol, which cannot be directly linked to Boom Beach. Thus, white listing or blocking scenarios for Boom Beach need to include the Supercell protocol in order to perform efficiently.
BlueJeans	BlueJeans (iOS) BlueJeans (Android)	encrypted	Messaging	–
Burner	Burner (Android)	encrypted	VoiceoverIP	–
BitTorrent	BitTorrent (Android)	encrypted	Peer to Peer	Some bittorrent clients use Teredo tunnels for their file transfers. To properly detect these transfers it is necessary to enable the plain tunnel decapsulation.

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
CNN	CNN: Breaking US & World News (iOS) CNN: Breaking US & World News (Android)	video, apache, audio, encrypted, amazon-cloud	Web	The sports section of the CNN-app works on bleacherreport.net (especially the pictures). Therefore these flows are not detected as CNN. Last improved in dpisig_2019-10-11.
ConstantContact	Constant Contact (iOS) Constant Contact (Android)	encrypted	Business	–
CloudMe	CloudMe (iOS) CloudMe (Android)	audio, video, encrypted	File Transfer	–
ClashofClans	Clash of clans (iOS) Clash of clans (Android)	encrypted	Gaming	Gamekit correlation and helpshift flows are not correlated as clashofclans. Traffic generated due to connecting fb, twitter, insta, youtube are not correlated. Support flows are getting detected as Vendor Signature ‘Supercell’. Traffic correlation issues are being tracked with AN-46767.
ClashRoyale	Clash Royale (iOS) Clash Royale (Android)	encrypted	Gaming	In iOS, when watching news and esports from menu, it is detected as youtube. This flow is also not correlated as ClashRoyale. Traffic correlation issues are being tracked with AN-46767. Last improved in dpisig_2019-08-02.
CNTV	CNTV (iOS)	video, apache, audio	Streaming	–
DeNA	Godus (iOS) Godus (Android) Final Fantasy Record Keeper (iOS) Final Fantasy Record Keeper (Android)	encrypted, apache, video, amazon-cloud, google	Gaming	Last improved in dpisig_2019-10-11.

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
xCum-com	–	video, encrypted	Streaming	xCUM-com is using different streaming platforms and CDN services. These flows are not classified as xCUM-com. Also the xCum-com application produces high amounts of advertisement and background traffic. Therefore, the amount of traffic that is classified as xCum-com can be quite low in comparison. Currently, the attribute video is only set for HTTP traffic. Therefore, blocking of video application attribute is not possible. Last improved in dpisig_2019-10-11.
Demand5	–	stream, web, video, apache, google	Deprecated	–
Dailymotion	DailyMotion (iOS) DailyMotion (Android)	video, encrypted, apache, audio	Streaming	–
Deezer	Deezer Music (iOS) Deezer Music Player (Android)	audio, video, stream, apache, encrypted	Streaming	The traffic generated for Support option is getting detected as zendesk(android) and amazon cloud (ios). These flows are not correlated to deezer. Traffic correlation issues are being tracked with AN-46767.
CtiTV	CtiTV (iOS) CtiTV (Android)	video, apache	Streaming	The smartphone app uses youtube streams to provide videos.
DAZN	DAZN Live Sports Streaming (iOS) DAZN Sport Live Streaming (Android)	encrypted, amazon-cloud	Streaming	–
CrimeCity	Crime City (iOS) Crime City (Android)	apache	Gaming	–
DriveLock	DriveLock Privately (iOS)	encrypted	Business	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
DigicelPlayGo	Digicel PlayGo (iOS) PlayGo (Android)	encrypted	Streaming	–
DocuSign	DocuSign (iOS) DocuSign (Android)	encrypted	Business	–
DigicelTopUp	Digicel Top Up (iOS) Digicel Top Up (Android)	encrypted	Business	–
DigicelMusic	d'Music (iOS) d'Music (Android)	encrypted, apache, amazon-cloud	Streaming	Digicel Music uses Amazon Cloud as content provider. Therefore, streaming and download content will be classified as Amazon Cloud.
Dropbox	Dropbox (iOS) Dropbox (Android)	encrypted, file-transfer	File Transfer	–
Dubsmash	Dubsmash (iOS) Dubsmash (Android)	encrypted, amazon-cloud	Social	–
Dofus	Dofus Touch (iOS) Dofus Touch (Android)	–	Gaming	–
iHeartRadio	iHeartRadio (iOS) iHeartRadio (Android)	audio, encrypted, apache	Streaming	–
IcFlix	ICFLIX (iOS) ICFLIX (Android)	encrypted, apache, video, stream	Streaming	–
Grooveshark	Grooveshark Grooveshark	audio	Deprecated	–
Here	Here WEGo (iOS) Here WEGo (Android)	encrypted, apache	Web	–
eBay	eBay (iOS) eBay (Android)	apache, encrypted, google	Business	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
EasyTaxi	Easy, a Cabify app (iOS) Easy, a Cabify app (Android) Easy Taxi (iOS)	encrypted	Social	The usage of this application is limited to Brazilia and mostly Middle- and South America as well as the Middle East.
ESPN	ESPN (iOS) ESPN (Android)	encrypted, apache, video, audio, amazon-cloud	Streaming	Disneys CDNs produce much traffic containing ESPN but will not be accounted. Last improved in dpisig_2019-10-11.
Duokan	Duokan (iOS)	encrypted, apache	Business	–
FaceTime	FaceTime (iOS)	encrypted, audio, video	VoiceoverIP	Last improved in dpisig_2019-10-11.
Evernote	EverNote (iOS) EverNote (Android)	encrypted, apache	Web	–
Endomondo	Endomondo (iOS) Endomondo (Android)	encrypted, apache, google	Social	–
FarmHero	Farm Heroes Saga (iOS) Farm Heroes Saga (Android)	encrypted	Gaming	–
Hotmail	–	apache, web, chat, encrypted, google	Deprecated	–
FETnet	FETnet (iOS) FETnet (Android)	encrypted, apache	Streaming	–
FubonE-Broker	Fubon E-Broker (iOS) Fubon E-Broker (Android)	encrypted	Web	–
file-upload-net	File Upload .net Uploader (Android)	ddl, apache, video	Sharehostin g	–
FileCloud	File cloud (iOS) File cloud (Android)	–	–	–
filefront-com	filefront-com	–	Deprecated	–
filesafe-to	–	–	Deprecated	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
Freenet	Freenet (iOS) Freenet (Android)	–	PeertoPeer	–
FuzeMeeting	Fuze Mobile (iOS) Fuze Mobile (Android)	encrypted, amazon-cloud	Business	–
fshare-vn	Fshare (iOS) Fshare (Android)	–	Sharehosting	–
Friendster	Friendster	Deprecated	–	–
Funshion	Funshion (iOS)	video, apache, audio	Streaming	Funshion uses the streaming service of Youku. Thus, the application Youku can be detected within Funshion traffic.
GameofWar	Game of War (iOS) Game of War (Android)	encrypted, audio	Gaming	–
Foursquare	Foursquare City Guide (iOS) Foursquare City Guide (Android)	encrypted	Social	–
GoogleAllo	Google Allo (iOS) Google Allo (Android)	encrypted, google	Messaging	–
GoogleAnalytics	Google Analytics (iOS) Google Analytics (Android)	encrypted, google, apache, blackberry	Business	Parts of the Google Analytics traffic is detected as Google APIs or Google Shared Services with the attribute encrypted. Therefore, it is not always possible to block Google Analytics.
IDrive	Idrive Online Backup (iOS) Idrive Online Backup (Android)	encrypted	File Transfer	SSL Session ID Tracking (SIT) is required for the detection to work properly. In a blocking scenario SIT is not necessary because the client won't be able to obtain a session ID.
Goober	Goober	–	Deprecated	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
GoogleDrive	Google Drive (iOS) Google Drive (Android)	google, encrypted	File Transfer	Most of the Google Drive traffic is detected as Google Shared Services with the attribute encrypted. The Google Shared Services traffic is higher on mobile devices. In these cases, it is not possible to block Google Drive.
Giphy	Giphy (iOS) Giphy (Android)	encrypted, video	Social	–
GoogleHangouts	Hangouts (iOS) Google Hangouts (Android)	file-transfer, google, video, audio, encrypted	Messaging	A change of the application attributes audio to video for one flow is possible. Video connections can be established for a short amount of time (10 to 15 seconds) until they are classified as 'video'. Google Hangouts uses generic functions of the Google platform for login and messaging, which are used by other Google applications too. Therefore, it is not possible to differentiate them in each case. The affected traffic will mostly be classified with attributes encrypted and Google. Especially on iOS and Browser, messaging and sending files will mostly be classified as Google Services. On Android, incoming file-transfer will be classified as Google Services. Also, blocking of chat and file-transfer is not possible because in blocking scenarios, Google Hangouts uses generic patterns for its traffic.
GoogleMaps	Google Maps (iOS) Google Maps (Android)	encrypted, google	Business	Most of the Google Maps traffic is detected as Google Shared Services with the attribute encrypted. The Google Shared Services traffic is higher on mobile devices. In these cases, it is not possible to block Google Maps.

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
GoogleEarth	Google Earth (iOS) Google Earth (Android)	encrypted, google, apache	Business	Most of the Google Earth traffic is detected as Google Shared Services with the attribute encrypted. The Google Shared Services traffic is higher on mobile devices. In this case it is not possible to block Google Earth. Google Street View is also a feature of Google Earth and will be classified as Google Earth.
GoogleCloud	Google Cloud Console (iOS) Cloud Console (Android)	encrypted, google	Web	Most of the Google Cloud traffic is detected as Google Shared Services with the attribute encrypted. The Google Shared Services traffic is higher on mobile devices. In this case it is not possible to block Google Cloud.
GoogleDocs	Google Docs (iOS) Google Docs (Android)	encrypted, google, apache	Business	Most of the Google Docs traffic is detected as Google Shared Services with the attribute encrypted. The Google Shared Services traffic is higher on mobile devices. In this case, it is not possible to block Google Docs
GoogleMail	Gmail (iOS) Gmail (Android)	google, encrypted, mail, apache, audio	Mail	Most of the Google Mail traffic is detected as Google Shared Services with the attribute encrypted. The Google Shared Services traffic is higher on mobile devices. In this case, it is not possible to block Google Mail.
HikeMessenger	Hike Messenger (iOS) Hike Messenger (Android)	blackberry, encrypted, video	Messaging	If the detection did not work and the packets are tagged as SSL the test should be run with the session ID tracker (SIT). For more information about SIT read the PACE integration manual.
Half-Life2	Half-Life 2 (Android)	apache, audio, video	Gaming	The most gameplay traffic will be classified as application Source Engine.

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
HiDrive	HiDrive (iOS) HiDrive (Android)	encrypted	File Transfer	All access methods of HiDrive need to be tested, these are Browser, VPN, WebDAV, SMB, App. SSL Session ID Tracking (SIT) may need to be enabled for Browser and WebDAV tests. Between each test the browser should be restarted so existing SSL Session ID's are removed. The login to non HiDrive services on the Strato page should not be blocked. Browser - Log onto the HiDrive website and do file transfers in the browser VPN - Option in the windows client WebDAV - Option in the windows client SMB - Mount HiDrive directly with SMB App - Mobile phone/tablet apps.
HalfbrickStudios	halfbrick-studios (iOS) halfbrick-studios (Android) Fruit Ninja (iOS) Fruit Ninja (Android)	encrypted	Gaming	It is not possible to successfully block the games provided by Halfbrick Studios because they are mostly offline games. Also, the detection rate might be low because of lots of advertisement traffic.
Grindr	Grindr (iOS) Grindr (Android)	encrypted, amazon-cloud, video, apache, google	Social	Last improved in dpisig_2019-10-11.
HamiVideo	Hami Video (Android)	apache	Streaming	–
hi5	Hi5 (Android)	encrypted, apache, audio	Social	hi5 was acquired by Tagged and many Tagged flows appear within hi5. Last improved in dpisig_2019-10-11.
GooglePlayMusic	Google Play Music (iOS) Google Play Music (Android)	audio, google, encrypted, video	Streaming	Most of the Google Play Music traffic is detected as Google Shared Services with the attribute encrypted. The Google Shared Services traffic is higher on mobile devices. In this case, it is not possible to block Google Play Music.
Hightail	HighTail (iOS) HighTail (Android)	encrypted	Business	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
GooglePlay	Google Play (Android) Google Play Games (Android)	encrypted, google	Business	Most of the Google Play traffic is detected as Google Shared Services with the attribute encrypted. The Google Shared Services traffic is higher on mobile devices. In this case it is not possible to block Google Play.
GooglePlus	Google + for G suite (iOS) Google + for G suite (Android)	google, encrypted	Social	Most of the Google Plus traffic is detected as Google Shared Services with the attribute encrypted. The Google Shared Services traffic is higher on mobile devices. In this case it is not possible to block Google Plus.
GoToMyPC	Go To My PC (iOS) Go To My PC (Android)	encrypted, apache	RemoteControl	–
Hulu	Hulu (iOS) Hulu (Android)	encrypted, video, stream, apache, audio, google, chat	Streaming	–
Huddle	Huddle (iOS) Huddle (Android)	encrypted, amazon-cloud	Business	Huddle uses additional CDNs like garenanow.com or amazon-cloud and they will be classified as amazon-cloud or new-relic.
GoToMeeting	Go To Meeting (iOS) Go To Meeting (Android)	encrypted, apache	Conference	–
HayDay	Hay Day (iOS) Hay Day (Android)	–	Gaming	–
GoogleSharedServices	Google Shared Services/google-services Google Shared Services/google-services	google, encrypted, video, apache, audio, blackberry, mail	Web	Google uses DNS canonical names (CNAME) to resolve different domains with the same IP address. In such a case an established connection to one service (example: Gmail) can be reused for request to other services (e.g. Google Cloud). This mechanism influences blocking scenarios of certain services so that a packet blocking cannot be done for each request. Last improved in dpisig_2019-08-30.

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
Hyread	HyRead Library (iOS) HyRead Library (Android) HyRead 3 (Android) HyRead (Android)	encrypted, apache	Web	–
magicJack	magicApp Calling & Messaging (iOS) magicApp Calling & Messaging (Android)	encrypted, apache, audio	VoiceoverIP	–
WindowsAzure	Windows Azure (iOS) Microsoft Azure (Android)	encrypted	FileTransfer	Last improved in dpisig_2019-10-11.
ModernWar	Modern War by GREE (Android)	apache, amazon-cloud	Gaming	–
Vippie	Vippie (Android)	audio, video, encrypted, google	VoiceoverIP	SSL Session ID Tracking (SIT) is required for the classification to work properly.
LinkedIn	Linkedin (iOS) Linkedin (Android)	encrypted, apache	Social	Last improved in dpisig_2019-10-18.
ZohoWorkOnline	Zoho Workdrive (Android)	encrypted, apache	Business	–
Flickr	Flickr (iOS) Flickr (Android)	apache, encrypted	Social	The web-page and the applications use a generic yahoo login server which will not be detected as Flickr. Because of this, the login should work during a block test, but fetched content (i.e. pictures) will be blocked.
Gadu-Gadu	GG (Android)	video, audio, file-transfer, encrypted, apache	Messaging	Only file transfers with a minimum throughput of 50 kB/s and a duration longer than 1 second can be detected.
filecloud-com	Filecloud (Android)	Deprecated		–
mediafire-com	MediaFire (iOS) MediaFire (Android)	ddl, apache, video, encrypted	Sharehosting	Last improved in dpisig_2019-10-11.

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
Iqiyi	QIYI (iOS) IQIYI (Android)	apache, video, audio, encrypted	Streaming	–
Life-Ru	Life.ru (iOS) Life.ru (Android)	encrypted	Web	–
WorldOfTanks	World Of Tanks (iOS) World Of Tanks (Android) World of Tanks Assistant (Android)	encrypted	Gaming	This classification currently only indicates World of Tank related traffic. Proprietary login or gaming traffic is not covered. Please be aware that blocking is not possible.
EagleVPN	Eagle Vpn (Android)	encrypted	Tunnel	In a blocking scenario it is not guaranteed to block this application completely. It can happen that the underlying L7 protocol Shadowsocks cannot be classified, because of the usage of TCP mode with port 443. More information can be found inside the limitations section of the protocol Shadowsocks.
BriaMobile	Bria Mobile (iOS) Bria Mobile (Android)	audio, video	Business	Last improved in dpisig_2019-07-19.
BriaTeams	Bria Teams (iOS) Bria Teams (Android)	audio, video	Business	–
VPNGate	VPN Download (Android) EasyOVPN - Plugin for open vpn (Android) Free VPN (Android) Free VPN Proxy (Android)	encrypted	Tunnel	–
live-com	–	Deprecated	–	–
icq_webchat	icq_webchat	Deprecated	–	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
Last.fm	Last.fm (iOS) Last.fm (Android)	audio, encrypted, apache, stream	Streaming	LastFM embeds external audio and video content on their website. The traffic of this embedded content is classified as the website it originates from. There is a low detection rate for the audioscrobbler, because it uses other services which are not part of LastFM. Last improved in dpisig_2019-10-11.
Mega	Mega	Deprecated	—	—
Jingdong	Jingdong (iOS) Jingdong (Android)	video, encrypted, apache, audio	Business	—
MagV	Magv (iOS) Magv (Android)	encrypted	Business	—
LeTV	LeTV (iOS)	stream, web, video, audio, apache, encrypted	Streaming	—
MyFitnessPal	MyFitnessPal (iOS) Calorie Counter (Android)	encrypted	Social	MyFitnessPal uses Brightcove as CDN to distribute video content, therefore a lot of traffic can be classified as Brightcove.
Sohu	Sohu (iOS) Sohu video - HD play headline film and television (iOS) Sohu (Android)	video, apache, encrypted	Web	—
Bolt	Bolt browser Documents (iOS) Bolt Speed Browser (Android)	apache, video	Mobile	—

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
IMO	imo video calls and chat (iOS) imo video calls and chat (Android)	encrypted, google, audio, video	Messaging	IMO-client reacts to blocking of audio- and video- stream related UDP-flows with shifting the traffic behavior. Blocking scenarios of dedicated audio- or video-attributes are therefore not supported. While opening the chat-functionality of the app or transferring files, IMO produces SSL traffic without handshake. This traffic can not be classified as IMO. Because of this limitation, also blocking of chat- and file-transfer-traffic is not possible. If you also block the related SSL flows without handshake, IMO will start using generic services provided by the used Smartphone. For Android this can be Google Firebase Cloud Messaging. For iOS the service can be Apple Push Notification Service.
KikMessenger	kik (iOS) kik (Android)	encrypted, apache, amazon-cloud, google, video	Messaging	On iOS Kik Messenger uses cloud services to deliver parts of its traffic.
Kakaotalk	KakaoTalk (iOS) KakaoTalk (Android)	encrypted, apache, video	Messaging	KakaoTalk uses generic functions of the Kakao platform for login and messaging, which are used by other Kakao applications too. The affected traffic will be detected as Kakao
KollectBook	Kollect-Book (iOS) Hami Book (Android)	encrypted, apache, video	Web	—
Kono	Kono (iOS) Kono (Android)	encrypted, amazon-cloud	Social	—
iLeader	ileader (iOS) ileader (Android)	encrypted	Business	—
LineRangers	Line Rangers (iOS) Line Rangers (Android)	encrypted, apache, audio, video	Gaming	—

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
PlayStationNetwork	Play Station App (Android)	amazon-cloud, apache, encrypted, file-transfer, google, stream, video	Gaming	The file-transfer application attribute indicates the downloads of an System or Game Update. Game updates that runs over playstation CDN service are covered. It is possible that publishers using different CDN services. Last improved in dpisig_2019-10-11.
LeagueofLegends	League of Legends Friends (iOS) League Friends (Android)	encrypted, apache	Gaming	–
Odnoklassniki	OK (iOS) OK (Android)	encrypted, apache, audio, video, google, mail	Social	–
NortonSecurity	Norton Mobile Security (iOS) Norton Mobile Security & Antivirus Protection (Android)	encrypted, apache	Business	–
Good	Blackberry Work (iOS) Blackberry Work (Android)	encrypted	Business	–
Signal	Signal Private Messenger (iOS) Signal Private Messenger (Android)	encrypted, amazon-cloud, file-transfer	Messaging	–
KanKan	Kankan (iOS)	stream, encrypted, video, apache	Streaming	For Android, the video-content is provided by a generic Chinese host and is therefore not detected as KanKan. Also, some parts of the traffic will be detected as Thunder, because this is the provider of this application.
Mail-Ru	Mail.Ru Group 20 лет (iOS) Mail.ru – Email App (iOS) Mail.ru - Email App (Android) Cloud Mail.ru (iOS) Cloud Mail.ru (Android) Mail.Ru Agent (iOS)	encrypted, apache, file-transfer, audio, video	Mail	Although ICQ is owned by Mail-Ru, it has it's own patterns and is mostly not classified as Mail-Ru. The same is true for Agent, a Russian copy of ICQ. The e-mail app of Mail-Ru uses Facebook for some generic traffic. Therefore a block of this app will not be successful.

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
Ivi-Ru	ivi (iOS)	stream, video, encrypted	Streaming	–
Tinder	Tinder (iOS) Tinder (Android)	encrypted	Social	Tinder is using Facebook, so there is related traffic
Tenor	GIF Keyboard (iOS) GIF Keyboard (Android)	encrypted, video	Messaging	Last improved in dpisig_2019-08-02
ChannelOne	Channel One (Android)	stream, video, encrypted	Streaming	Watching videos may use cdn.ngenix.net as provider, which cannot be classified for this app clearly.
Imgur	Imgur (iOS) Imgur (Android)	encrypted, video	Social	The video attribute classification for encrypted traffic is based on throughput calculation. Throttling or other throughput influencing methods may result in preventing classification of video attribute. On iOS and Browser, blocking the attribute video leads to the case, that no specific content can be accessed. Last improved in dpisig_2019-10-11.
OneDrive	Microsoft Onedrive (iOS) Microsoft Onedrive (Android)	encrypted, file-transfer	File Transfer	Blocking OneDrive results in also blocking the OneDrive website because files may be up and downloaded from there as well. File transfers in other MSN services may no longer work when OneDrive is blocked as some of them use OneDrive for this. The OneDrive Android app will still show new files in the cloud even when blocked but it won't be able to download them. OneDrive uses generic Microsoft services which are classified as Microsoft service.
LogMeIn	Logmein (iOS) Logmein (Android)	encrypted	RemoteControl	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
MobileStrike	Mobile Strike (iOS) Mobile Strike (Android)	encrypted	Gaming	–
KingofAvalon	King Of Avalon (iOS) King Of Avalon (Android)	amazon-cloud	Gaming	–
Lineage2	Lineage 2 (iOS) Lineage 2 (Android)	encrypted	Gaming	Only game server traffic is classified as the login protocol causes too little traffic.
LifeChurch	Life.Church (iOS) Life.Church (Android)	encrypted, amazon-cloud	Social	–
LearningHub-online	Learning Hub (Android)	encrypted, apache, video	Business	–
GoogleAds	Google Ads (Android)	encrypted, google, apache, video, blackberry, audio	Advertisement and Analytic Services	–
SteganosServices	Stagenos Mobile Privacy (Android)	encrypted	Business	–
Ren-Tv	Cosmos3D : Ren tv (Android)	encrypted, video	Web	–
Brightcove	Brightcove	video, encrypted, web, apache	Business	The video attribute classification for encrypted traffic is based on throughput calculation. Throttling or other throughput influencing methods may result in preventing classification of video attribute. Last improved in dpisig_2019-10-11.
Sportbox-Ru	–	encrypted, video	Web	Sportbox-Ru uses the Russian search machine Yandex to provide some information, so a significant amount of traffic while using Sportbox-Ru will be classified as Yandex. Last improved in dpisig_2019-10-11.
Ookla	Ookla	encrypted	Network Management	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
AnchorFreeServices	Hexatech Unlimited VPN Proxy (iOS) VPN Proxy by Hexatech (Android) Hotspot Shield VPN & wifi Proxy (iOS) Hotspot Shield Free VPN Proxy & Wi-Fi Security (Android) Best VPN Proxy Betternet (iOS) VPNFree (Android) TouchVPN (iOS) TouchVPN (Android) VPNInTouch (iOS) VPNInTouch (Android) Just VPN (iOS) Just VPN (Android) VPN 360 - Unlimited VPN Proxy (iOS)	encrypted	Tunnel	One of the applications which are covered by AnchorFree Services is Betternet. Betternet uses multiple services for their traffic. For example, Domain Name Fronting (DNF), IPsec or OpenVPN. For most cases it is not possible to block Betternet. For the DNF case we cannot ensure a full classification.
SecureVPN	SecureVPN Unlimited Privacy (iOS) SecureVPN Free Unlimited Privacy (Android)	encrypted	Tunnel	Last improved in dpisig_2019-10-11.
ActmobileServices	Dash VPN (iOS) Dash VPN (Android) Free VPN by FreeVPN.org (iOS) Free VPN by FreeVPN.org (Android)	encrypted	Tunnel	Last improved in dpisig_2019-10-11.

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
Gameloft-Services	Minion Rush (iOS) Minion Rush (Android) Asphalt 8: Airborne (iOS) Asphalt 8: Airborne (Android) Asphalt9: Legends (iOS) Asphalt9: Legends (Android) Modern Combat 5 (iOS) Modern Combat 5 (Android) Dungeon Hunter 5 (iOS) Dungeon Hunter 5 (Android) N.O.V.A. Legacy (iOS) N.O.V.A. Legacy (Android) Order & Chaos Online (iOS) Disney Magic Kingdoms (Android) Dungeon Hunter Champions (Android) Dead Rivals - Zombie MMO (Android) Disney Magic Kingdoms (iOS) Dungeon Hunter Champions (iOS) Dead Rivals - Zombie MMO (iOS)	encrypted, apache	Gaming	Dungeon Hunter Champions: Linking FB account / google account is detected as FB or google account and not as dungeon hunter (gameloft). In addition, game provides access to fb/twitter/youtube pages to see the game related news and posts. These flows are also not correlated to dungeon hunter. Ads traffic (Vungle, unity etc and youtube ads) are not correlated to dungeon hunter. There is no strong enough pattern available to separate the traffic of this application from generic gameloft-services traffic. Traffic correlation issues are being tracked with AN-46767. Disney Magic Kingdom: The background traffic when launching the game and Ads traffic (Vungle, unity etc and youtube ads) are not correlated to disney magic kingdom. There is no strong enough pattern available to separate the traffic of this application from generic gameloft-services traffic. For correlation issue we are tracking in AN-46767. Last improved in dpisig_2019-10-11.
Asphalt9	Asphalt9: Legends (iOS) Asphalt9: Legends (Android)	encrypted	Gaming	This application produces a lot of additional advertisement traffic while starting, especially by Unity and Stackpath Cloud. Therefore, the basic detection rate is low in comparison to the caused traffic. Also, parts of the traffic will be classified as Gameloft Services
Shadowsocks	Shadowsocks (Android) Shadowsocks Free Account (Android)	encrypted	Tunnel	—

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
Smule	Smule (iOS) Smule (Android)	encrypted, video, audio, amazon-cloud	Social	Magic Piano has a very low detection rate, due to ad background traffic. The other apps have strongly varying detection rates, also due to ad background, but blocking works for all of them.
LoopPacificLocalNews	Loop Pacific (Android)	apache, amazon-cloud	Web	–
LoopCaribbeanLocalNews	Loop - Caribbean Local News (Android)	encrypted, apache	Web	–
RackspaceCloud	Rackspace Mobile (Android)	encrypted	Business	–
MeeticGroup	Meetic Chat (iOS) NEU.DE (Android) Meetic - Flirt et Rencontre (iOS)	encrypted, amazon-cloud	Social	–
YoukuTudou	youku-tudou (iOS) Youku (Android)	video, encrypted, apache, audio	Streaming	–
Cloudflare	1.1.1.1 : Faster & Safer Internet (Android)	encrypted	Business	Last improved in dpisig_2019-08-02.
StarWarsGalaxyofHeroes	Star Wars : Galaxy of heroes (iOS) Star Wars : Galaxy of heroes (Android)	encrypted	Gaming	Due to the use of shared cloud services only a small portion of the traffic is classified as starwars-goh.
SimCityBuildIt	SimCity Buildit (iOS) SimCity Buildit (Android)	apache, audio	Gaming	The majority of the game traffic are shared cloud services. As such the amount of traffic classified as SimCity BuildIt may be relatively low.
NickelodeonPlay	Nickelodeon Play (iOS) Nickelodeon Play (Android)	encrypted	Streaming	Only small amounts of traffic are classified as the majority of traffic uses external streaming providers. Last improved in dpisig_2019-10-11.
Trulia	Trulia Real Estate (iOS) Trulia Real Estate (Android)	encrypted, amazon-cloud, apache	Business	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
Lyft	Lyft (iOS) Lyft (Android)	encrypted, amazon-cloud	Business	–
Tripadvisor	TripAdvisor (iOS) TripAdvisor (Android)	encrypted, apache	Business	–
TomTom	TomTom Go Navigation (iOS) TomTom GPS Navigation (Android) TomTom VIO (iOS) TomTom MyDrive (iOS)	encrypted, apache	Business	–
MyDigicel	My Digicel (iOS) My Digicel (Android)	encrypted, apache	Web	–
Synology	Synology Drive (Android)	encrypted	File Transfer	–
TuneInRadio	TuneIn (iOS) TuneIn (Android)	encrypted, audio, apache, video, amazon-cloud	Streaming	Audio streaming is classified as Shoutcast/Icecast. This will not be classified as TuneIn Radio.
Shazam	Shazam (iOS) Shazam (Android)	apache, encrypted	Social	In a blocking scenario it is not possible to block the starting page of this application. This traffic is offline traffic. Last improved in dpisig_2019-10-11.
Psiphon	Psiphon Pro (Android)	encrypted	Tunnel	Only proprietary traffic is covered by this classification. SSH, HTTP and ipsec psiphon-related traffic will not be detected. Therefore, no blocking is possible.
Kinopoisk	Kinopoisk (iOS)	encrypted	Web	–
SkyGo	Sky Go (iOS) sky-go (iOS) Sky Go (Android)	stream, video, apache, encrypted	Streaming	–
eDisk	eDisk (Android)	ddl, apache	Sharehosting	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
Yunfile	Yunfile (iOS)	ddl, apache	Sharehosting	The application Yunfile contains a lot of advertisement and external traffic which influences the detection rate.
Prisma	Prisma Photo Editor (iOS) Prisma Photo Editor (Android)	encrypted	Social	–
Rutube	RutubeSeries (iOS) Rutube (iOS) Rutube (Android)	encrypted, video, audio	Streaming	–
PokemonGo	Pokemon Go (iOS) Pokemon Go (Android)	encrypted, google	Gaming	–
Slack	Slack (iOS) Slack (Android)	encrypted, amazon-cloud	Business	–
ZenDesk	zendesk Support (iOS) Zendesk Chat (iOS) Zendesk Sell (iOS) zendesk Chat (Android) Zendesk Support (Android) Zendesk Sell (Android)	encrypted, amazon-cloud	Business	Last improved in dpisig_2019-10-11.
NewRelic	New Relic (iOS) New Relic Android App (Android)	encrypted, web, apache, video	Business	–
SurveyMonkey	SurveyMonkey (iOS) SurveyMonkey (Android)	encrypted	Business	SurveyMonkey opens lots of connections to 3rd party content, like facebook, resulting in a classification rate of 30
iTunesU	iTunes U (iOS)	apache, encrypted	Social	Browsing in iTunes U store is also classified as iTunes or encrypted SSL when using a generic Apple CDN service.
PochtaRossii	Pochta Rossii (iOS) Pochta Rossii (Android)	encrypted	Business	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
TMall	tmall (iOS) tmall (Android)	encrypted, video	Business	–
VK	VK - social network and calls (iOS) VK - social network (Android)	audio, encrypted, apache, video	Social	Last improved in dpisig_2019-08-30.
iCloud	iCloud Drive (iOS)	encrypted	File Transfer	–
Zynga	–	audio, encrypted, amazon-cloud, apache, video	Gaming	The video attribute detection for Zynga is based on HTTP and does not work for TLS traffic.
SugarSync	SugarSync (iOS) SugarSync (Android)	video, encrypted, apache	File Transfer	–
Wickr	Wickr Me (iOS) Wickr Me (Android)	encrypted	Messaging	–
TigerText	TigerConnect (iOS) TigerConnect (Android)	encrypted	Messaging	–
Vibe	Vibe (iOS)	encrypted, amazon-cloud, apache	Messaging	–
Voxer	Voxer Walkie Talkie Messenger (iOS) Voxer Walkie Talkie Messenger (Android)	encrypted	Messaging	Blocking Voxer results in also blocking parts of the website.
Spread	Spread (iOS) Webinar (iOS) Webinar (Android)	encrypted	Conference	–
textPlus	textPlus (iOS) textPlus (Android)	audio, video, encrypted, amazon-cloud, apache	VoiceoverIP	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
Snapchat	Snapchat (iOS) Snapchat (Android)	encrypted, google, audio, video	Messaging	On Android it is not possible to distinguish between file-transfer, audio and video calls. Audio and video traffic are transported within the same flow, so if you start a video call and close the camera within the same call the traffic will still be classified as video. Last improved in dpisig_2019-10-11.
MyMusic	My Music My Music (Android)	audio, video, encrypted, apache	Streaming	–
Truphone	My Truphone (iOS) Truphone SIM Data Settings (Android)	audio, encrypted	VoiceoverIP	–
XBOXgamingconsole	xbox	apache, encrypted, video	Gaming	Last improved in dpisig_2019-10-18.
Yelp	Yelp For Business Owners (iOS) Yelp - Food & Services Around Me (iOS) Yelp (Android) Yelp for Business Owners (Android)	encrypted, apache	Social	–
Spideroak	SpiderOakONE (iOS) SpiderOak One (Android)	encrypted	File Transfer	–
Tumblr	Tumblr (iOS) Tumblr (Android)	video, encrypted, apache	Social	Last improved in dpisig_2019-10-11.
XING	Xing (iOS) Xing (Android)	encrypted, apache	Social	–
Vimeo	Vimeo (iOS) Vimeo (Android)	web, video, encrypted, apache, audio	Streaming	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
Salesforce	Salesforce (iOS) Salesforce (Android)	encrypted	Business	–
iTunesRadio	iTunes Radio	–	Streaming	–
SinaWeibo	Weibo (iOS) Weibo (Android) Weibo (Android)	encrypted, apache, video, audio	Social	–
Baidu	Baidu (iOS)	encrypted, apache, video	Web	Playing videos or music with Baidu is not detect as application Baidu. This traffic is detected as HTTP with subtype video, audio or media or at least as protocol QUICKTIME, FLASH or MPEG. In a block scenario it is not possible to search or play videos /music.
Taobao	Taobao (iOS) Taobao (Android)	video, encrypted, apache	Business	Last improved in dpisig_2019-08-30.
Watchever	Watchever (iOS)	video, encrypted, apache, google	Deprecated	–
Pinterest	pinterest (iOS) pinterest (Android)	audio, video, encrypted, google, apache, amazon-cloud, chat	Social	The video attribute classification for encrypted traffic is based on throughput calculation. Throttling or other throughput influencing methods may result in preventing classification of video attribute. Last improved in dpisig_2019-10-11.
TimelyTV	timely-tv (iOS) timely-tv (Android)	apache	Streaming	Some of the videos are hosted on YouTube. These videos are detected as application YouTube.
WorkoutTrainer	workout-trainer (iOS) workout-trainer (Android)	video, amazon-cloud, audio	Social	The block-test for Android does not work completely: videos are blocked but

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
MapMyRun	mapmyrun (iOS) mapmyrun (Android)	audio, encrypted, amazon-cloud, google	Social	The maps traffic of Map my Run depends on the used device and will be classified as SSL. Additionally, lots of ads and analytic traffic is caused by Map my Run, which is detected separately. Last improved in dpisig_2019-10-11.
Yahoo	Yahoo Search(iOS) Yahoo Mail (iOS) Yahoo Mail (Android) Yahoo Search (Android)	encrypted, audio, video, file-transfer, web, chat, mail, google blackberry	Messaging	Last improved in dpisig_2019-10-11.
vonage	vonage (Android) Vonage Mobile Call Video Text (Android) Vonage Mobile (iOS) Vonage Business Cloud (iOS)	–	Voice over IP	–
Vcast	vcast (Android)	–	Deprecated	–
sendspace-com	sendspace-com (Android)	ddl, encrypted, apache, video	Sharehosting	–
shragle-com	shragle-com (Android)	ddl, apache	Sharehosting	–
SopCast	sopcast (Android)	–	Streaming	–
Zattoo	zattoo (iOS) zattoo (Android)	encrypted, apache, video, audio, google	Streaming	Last improved in dpisig_2019-07-19.
QQLive	qqlive (Android)	–	Streaming	The goal of the QQLive detection is to detect the video traffic. The website should be accessible during a block test.
VNC	vnc (iOS) vnc (Android)	encrypted	Remote Control	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
Steam	Steam Mobile (iOS) Steam (Android)	encrypted, apache, video, google, audio	Gaming	Encrypted traffic from the steam website will also be detected as protocol Steam, because the applications use the website for community and shopping activities. Last improved in dpisig_2019-08-02.
Telnet	telnet (Android)	–	Remote Control	–
TOR	tor (Android)	encrypted	Tunnel	For traffic obfuscated with either obfs2 or obfs3 it is required that packets enter the detection in correct order. Testing may fail for obfs2 and obfs3 traffic when packets are out of order, verify the result by running CAAP or PAT on the trace file with PARO enabled.
WorldofWarcraft	worldofwarcraft (Android)	encrypted, apache, video	Gaming	–
Skinny	skinny (Android)	–	Voice over IP	–
Paltalk	paltalk (iOS) paltalk (Android)	encrypted, audio, video, file-transfer	Messaging	Last improved in dpisig_2019-08-30.
Pandora	pandora (iOS) pandora (Android)	apache, video, encrypted, audio, google	Streaming	Last improved in dpisig_2019-10-11.
TeamSpeak	teamspeak (iOS) teamspeak (Android)	–	Voice over IP	–
MapleStory	maplestory (Android)	apache	Gaming	–
Nimbuzz	nimbuzz (iOS)	blackberry, encrypted, apache, amazon-cloud, audio, google	Deprecated	–
CrossFire	crossfire (Android)	–	Gaming	–
Armagetron	armagetron (Android)	–	Gaming	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
IMplus	implus (Android)	encrypted, apache, audio, video, amazon-cloud, google	Messaging	Last improved in dpisig_2019-10-11.
UltraSurf	ultrasurf (Android)	encrypted, google	Tunnel	Running the Windows client of Ultrasurf without browsing cannot be safely classified and blocked, as Ultrasurf uses 3rd party servers for connection establishment. Last improved in dpisig_2019-08-30.
iPlayer	iPlayer (iOS)	apache, audio, video, google	Streaming	Last improved in dpisig_2019-10-11.
OperaMini	operamini (iOS) operamini (Android)	audio, apache, encrypted, google	Mobile	–
Webex	webex (iOS) webex (Android)	encrypted	Conference	–
eBuddy	ebuddy (Android)	audio, apache	Messaging	Last improved in dpisig_2019-10-11.
Wuala	wuala (iOS)	–	Deprecated	–
WebDAV	webdav (iOS) webdav (Android)	apache	Filetransfer	–
Audiogalaxy	audiogalaxy (iOS)	encrypted, audio	Streaming	–
ITV	itv Hub (iOS) itv Hub (Android)	stream, web, encrypted, audio, apache, google, video, amazon-cloud, chat	Streaming	Last improved in dpisig_2019-10-11.
Channel4oD	channel4od (Android)	encrypted, audio, video, apache	Streaming	–
Iperf	iperf (Android)	–	Network Management	UDP packets with less than 16 byte layer 7 payload are not detected as Iperf and remain as unknown traffic.

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
MicrosoftDynamicsNAV	msnav (Android)	encrypted, ntlm	Business	–
Destiny	destiny (Android)	apache, file-transfer	Gaming	It is possible that game updates will be detected as HTTP because they are not directly related to the gameplay.
VyprVPN	vyprvpn (iOS) vyprvpn (Android)	encrypted	Tunnel	VyprVPN can use different security and tunneling protocols like L2TP, PPTP, OpenVPN or IPSEC. In addition there is a modified OpenVPN 256-Bit mode called Chameleon. Using VyprVPN with the different protocols: - Android - OpenVPN is detect as VyprVPN with real protocol OpenVPN - Chameleon is detect as real protocol VyprVPN - Mac OSX - OpenVPN is detect as VyprVPN with real protocol OpenVPN - Chameleon is detect as real protocol VyprVPN - L2TP as L2TP and IPSEC - PPTP as PPTP and PPP - Windows - OpenVPN is detect as VyprVPN with real protocol OpenVPN - Chameleon is detect as real protocol VyprVPN - L2TP as L2TP and IPSEC - PPTP as PPTP and PPP - iOS - Detect as protocol IPSEC For all platforms the generic VyprVPN traffic is detect as VyprVPN.
Threema	threema (iOS) threema (Android)	encrypted	Messaging	The location search through the app is detect as Google or sometimes as unknown.
Twitch	twitch (iOS) twitch (Android)	encrypted	Streaming	–
Wowza	wowza (iOS) wowza (Android)	–	Streaming	A known issue is that it is not possible to classify RTMP flow from Wowza Streaming Engine as Wowza.
Runtastic	runtastic (iOS) runtastic (Android)	encrypted, apache, amazon-cloud, audio	Social	The maps traffic of Runtastic depends on the used device and will be classified as SSL. Additionally, lots of ads and analytic traffic is caused by Runtastic, which is detected separately.

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
Telegram	telegram (iOS) telegram (Android)	encrypted, apache, audio	Messaging	On iOS it is not possible to block file-transfer and chat traffic. In a blocking scenario, the speed of the transfer is reduced but it is not blocked completely. Last improved in dpisig_2019-10-11.
StreetVoice	streetvoice (iOS) streetvoice (Android)	amazon-cloud, audio, encrypted	Social	–
Skout	skout (iOS) skout (Android)	video, apache, google, audio, amazon-cloud, encrypted	Social	Skout uses content from other hosts this could cause a low detection rate.
PlayTales	playtales (iOS)	amazon-cloud	Gaming	–
PinkFong	pinkfong (iOS) pinkfong (iOS) pinkfong (iOS) pinkfong (iOS) pinkfong (iOS) pinkfong (iOS) pinkfong (Android)	audio, encrypted, amazon-cloud, video	Streaming	–
Moves	moves (iOS) moves (iOS)	encrypted	Deprecated	–
Syncplicity	syncplicity (iOS) syncplicity (Android)	encrypted	Filetransfer	–
YuantaWeb	yuanta-web (Android)	–	Business	–
HiNet	hinet (Android)	video, encrypted, apache, audio	Streaming	–
MyCard	mycard (Android)	encrypted	Streaming	–
Speakaboos	speakaboos (iOS) speakaboos (Android)	audio, video, encrypted, apache	Streaming	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
Taaze	taaze (iOS) taaze (Android)	encrypted, apache	Web	–
pububookbuffet	pubu-bookbuffet (iOS) pubu-bookbuffet (Android)	encrypted, amazon-cloud	Web	–
Mitake	mitake (iOS) mitake (iOS) mitake (iOS) mitake (Android)	–	Business	–
Qualys	qualys (Android)	encrypted, amazon-cloud	Business	–
IntraLinks	intralinks (Android)	encrypted	Business	–
Speedtest	speedtest (iOS) speedtest (Android)	apache, encrypted	Network Managemen t	–
Naver	naver (iOS) naver (Android)	audio, video, encrypted, apache,	Web	–
Periscope	periscope (iOS) periscope (Android)	encrypted, video, amazon-cloud	Streaming	Last improved in dpisig_2019-10-11.
PayEasy	payeasy (iOS) payeasy (Android)	encrypted, apache	Business	–
MyVideoTW	myvideo-tw (iOS)	apache, encrypted	Streaming	–
Supercell	Brawl Stars (iOS) Brawl Stars (Android)	encrypted	Gaming	–
Minecraft	minecraft (iOS) minecraft (Android)	–	Gaming	–
CitrixXenApp	Citrix Workspace (Android)	encrypted, ntlm	Business	Some traffic may be classified as Windows Azure because of the usage of these service for file storage.

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
Zalo	zalo (iOS) zalo (Android)	encrypted, video	Messaging	Sometimes it is not possible to block chat messages on Android devices.
Wikipedia	wikipedia (iOS) wikipedia (Android)	video, apache, encrypted	Web	–
Ustream	ustream (iOS) ustream (Android)	encrypted, apache, video, stream	Streaming	Last improved in dpisig_2019-10-11.
Uber	uber (iOS) uber (Android)	encrypted	Business	–
SoftbankOnline shop	softbank-onlineshop (Android)	video, apache, encrypted	Business	–
Slacker	slacker (iOS) slacker (Android)	audio, apache, encrypted, video	Streaming	The detection rate of Slacker in the Browser-version is quite low because of ad videos.
reddit	reddit (iOS) reddit (Android)	encrypted, video	Social	Only the mainpage-traffic will be detected as reddit, the content of the linked pages will be detected as the applications they belong to. Last improved in dpisig_2019-10-11.
PlantsvsZombies2	plants-vs-zombies-2 (iOS) plants-vs-zombies-2 (Android)	encrypted, apache, video, amazon-cloud	Gaming	While the game is loading there is shown an advertisement video, which is classified as QUICKTIME. It is not possible to block the application usage because the game can be played offline. If you play online, the traffic will mostly be classified as SSL with application Popcap.
PayPal	paypal (iOS) paypal (Android)	encrypted	Business	–
Niconico	niconico (iOS) niconico (Android)	video, apache, encrypted, audio	Streaming	Last improved in dpisig_2019-10-11.
NTTDocomo	docomo (Android)	encrypted, apache	Business	–
Waze	waze (iOS) waze (Android)	encrypted	Social	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
OutlookMobile	Microsoft Outlook (iOS) Microsoft Outlook (Android)	encrypted	Business	–
iFlix	Iflix (iOS) Inflix Discover Movies & TV Shows (Android) iflix (Android)	encrypted, stream	Streaming	–
iCrypt	icrypt (iOS)	encrypted	Messaging	–
AppleMaps	Maps (iOS)	encrypted	Business	Most of Apple Maps traffic is detected as Apple Geolocation
Mapbox	Mapbox (Android)	encrypted	Business	–
AppsFlyer	–	encrypted	Advertisement and Analytic Services	–
Adjust	–	encrypted	Advertisement and Analytic Services	–
GenieVPN	GenieVPN - Fast VPN Proxy (iOS)	encrypted	Tunnel	–
TunVPN	TunVPN Free VPN (Android)	encrypted	Tunnel	–
helpshift	–	encrypted	Messaging	The service platform helpshift may be used by many different mobile apps as part of the messaging system (like pubg).
FastVPNMaster	–	encrypted	Tunnel	–
VPNUnlimitedProxy	VPN - Unlimited Privacy & Security Proxy (iOS)	encrypted	Tunnel	–
HighSpeedVPN	High-speed VPN (Android)	encrypted	Tunnel	–
VPNMasterbyAndroidTeam	VPN Master - Free VPN (Android)	encrypted	Tunnel	–
Mixer	Mixer - Interactive Streaming (iOS) Mixer - Interactive Streaming (Android)	encrypted	Streaming	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
Douyu	Douyu	encrypted, audio, video	Streaming	There are no official apps for iOS and Android in the corresponding app stores available. However, you can download the apps from: https://www.douyu.com/client?tab=client#mobile
TencentGames	PUBG MOBILE (iOS) PUBG MOBILE (Android)	encrypted	Gaming	Tencent Games released some games which are based on the social media platform QQ, therefore QQ-traffic is expected in context of some games.
OpenSuperHotspotVPN	VPN-Open Super Hotspot Browser (iOS)	encrypted	Tunnel	–
VPNProxyMasterUnlimited	VPN - Proxy Master Unlimited (iOS)	encrypted	Tunnel	Currently, PACE does not support the Proxy Mode of VPN Proxy Master Unlimited.
iTunes	–	audio, ios, video, encrypted, apache, google, stream	Streaming	Browsing in iTunes is also classified as encrypted SSL when using a generic Apple CDN service. Last improved in dpisig_2019-10-18.
MagineTV	–	–	Deprecated	–
MySpace	–	audio, apache, video, encrypted	Social	–
RealMedia	–	–	Deprecated	–
UbuntuOne	–	–	Deprecated	–
Yuilop	–	–	Deprecated	–
iMessage	–	–	Deprecated	–
AmazonCloud	–	audio, web, video, encrypted, file-transfer, amazon-cloud, apache, google	Web	–
SUDAPhone	–	audio, encrypted	VoiceoverIP	–
SINATV	–		Deprecated	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
VoipSwitch	–	audio	VoiceoverIP	–
Skydrive	–		Deprecated	–
Orkut	–	google, encrypted	Deprecated	–
Soribada	–	audio, encrypted, apache	Streaming	–
enLegion	–	apache, google	Deprecated	–
LOVEFiLM	–	–	Deprecated	–
ICQWebchat	–	–	Deprecated	–
Xvideos	–	video, apache, audio, encrypted	Streaming	XNXX is using streaming servers of XVideos, it will get detected as XVideos. Last improved in dpisig_2019-10-11.
Youku	–	–	Deprecated	–
Renren	–	audio, encrypted, apache, video	Social	–
YouPorn	–	video, encrypted, apache	Streaming	–
PornHub	–	video, apache, encrypted	Streaming	–
Forfone	–	encrypted, audio, amazon-cloud	Deprecated	–
LiveJasmin	–	video, apache, encrypted	Streaming	–
IPMultimediaSubsystem	–	audio, video	Mobile	Currently only SIP based IMS traffic will be detected as application ims.
TTPod	–	audio, video	Streaming	To use the application, a chinese IP address is required.
Doook	–	–	Messaging	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
YikYak	–	encrypted	Deprecated	–
NortonAntivirus	–	–	Deprecated	–
Rdio	–	–	Deprecated	–
MagVKids	–	–	Business	Downloading MagV Kids magazines can be detected as MagV, because it is not possible to distinguish between both. The same behaviour occurs for loading pictures in the store.
DotaLegend	–	–	Gaming	–
XNXX	–	apache	Streaming	XNXX is using streaming servers of XVideos, so the video content and preview images will get detected as XVideos. When blocking the traffic from XVideos the XNXX site is also not possible because XVideos serves static content for the XNXX site.
KingofPirates	–	audio	Gaming	This application produces a lot of additional advertisement or flash traffic. Therefore the basic detection rate is low in comparison to the caused traffic.
Tudou	–	–	Deprecated	–
KaZaa/Fasttrack	–	audio, video	Streaming	–
Gnutella	–	encrypted, audio, video	PeertoPeer	–
DirectConnect	–	encrypted, apache	PeertoPeer	–
AppleJuice	–	–	PeertoPeer	–
Soulseek	–	–	PeertoPeer	–
Ares	–	–	PeertoPeer	Asymmetric detection is supported only for traffic from the client side.
Mute	–	–	PeertoPeer	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
MSN	–	video, encrypted, file-transfer, audio, apache, google	Messaging	The Hotmail webclient has the capability to chat via MSN. Blocking MSN also disables the "manage" and "preferences" functionality of Hotmail. In addition it is possible to post on a Facebook wall via the Hotmail account using the Facebook api. This is not detected because the post is send from within a running encrypted SSL flow.
Oscar	–	–	Deprecated	–
MPlus	–	encrypted, apache, google	Messaging	–
IRC	–	–	Deprecated	–
adrive-com	–	–	Deprecated	–
badongo-com	–	ddl, video	Sharehosting	–
badongo-net	–	–	Deprecated	–
bestsharing-com	–	ddl, apache	Sharehosting	–
bigfilez-com	–	–	Deprecated	–
chinamofile-com	–	–	Deprecated	–
data-hu	–	ddl, encrypted, video	Sharehosting	–
depositfiles-com	–	ddl, encrypted, video	Sharehosting	–
easy-share-com	–	ddl, video	Sharehosting	–
fast-load-net	–	–	Deprecated	–
fast-share-com	–	–	Deprecated	–
file-upload-to	–	–	Deprecated	–
filearchiv-ru	–	ddl	Sharehosting	–
filebase-to	–	–	Deprecated	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
filefactory-com	–	ddl, apache, video	Sharehosting	–
filehost-tv	–	–	Sharehosting	–
filepost-ru	–	ddl	Sharehosting	–
filer-net	–	ddl, video	Sharehosting	–
files-upload-com	–	–	Deprecated	–
files-to	–	ddl, apache, video	Sharehosting	–
filesend-net	–	ddl, apache, video	Sharehosting	–
fileupyours-com	–	ddl, apache, video	Sharehosting	–
ftp2share-com	–	–	Deprecated	–
gigasize-com	–	ddl, apache, video	Sharehosting	–
gigeshare-com	–	–	Deprecated	–
hostggg-com	–	–	Deprecated	–
hotfile-com	–	–	Deprecated	–
ifolder-ru	–	–	Deprecated	–
keepmyfile-com	–	–	Deprecated	–
leteckaposta-cz	–	ddl, apache	Sharehosting	–
live-share-com	–	–	Deprecated	–
livedepot-net	–	–	Deprecated	–
load-to	–	ddl, apache, video	Sharehosting	–
massmirror-com	–	–	Deprecated	–
megarotic-com	–	–	Deprecated	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
megashare-com	–	ddl, apache, video	Sharehosting	–
megashares-com	–	ddl, apache, video	Sharehosting	–
megaupload-com	–	ddl, apache, video	Sharehosting	–
mofile-com	–	–	Deprecated	–
mofile-net	–	–	Deprecated	–
mytempdir-com	–	–	Deprecated	–
netload-in	–	ddl	Sharehosting	–
odsiebie-najlepsze-net	–	–	Sharehosting	–
quicksharing-com	–	–	Deprecated	–
rapidshare-com	–	ddl, video	Sharehosting	–
rapidshare-de	–	–	Deprecated	–
rapidupload-com	–	–	Deprecated	–
savefile-com	–	ddl, apache, video	Sharehosting	–
sendmefile-com	–	–	Deprecated	–
sharebase-to	–	–	Deprecated	–
sharebee-com	–	–	Deprecated	–
sharebig-com	–	–	Deprecated	–
sharebigfile-com	–	ddl, apache, video	Sharehosting	–
sharebigfile-com	–	–	Deprecated	–
simpleupload-net	–	ddl, apache, video	Sharehosting	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
speedshare-org	–	ddl, apache, video	Sharehosting	–
turboupload-com	–	–	Sharehosting	–
ultrashare-de	–	–	Deprecated	–
up-file-com	–	–	Sharehosting	–
uploaded-to	–	–	Deprecated	–
uploading-com	–	ddl, apache, video	Sharehosting	–
uploadingit-com	–	ddl, apache, video	Sharehosting	–
uploadpower-com	–	–	Deprecated	–
uploadyourfiles-de	–	–	Deprecated	–
vip-file-com	–	–	Deprecated	–
wiiupload-net	–	–	Deprecated	–
yourfilehost-com	–	ddl, apache, video	Sharehosting	–
yourfiles-biz	–	–	Deprecated	–
youtube-com	–	–	Deprecated	–
zshare-net	–	ddl, video	Sharehosting	–
4shared-vn	–	–	Deprecated	–
mega-1280-com	–	–	Deprecated	–
megashare-vn	–	–	Deprecated	–
fileserv-com	–	ddl, video	Sharehosting	–
filesonic-com	–	ddl, video	Sharehosting	–
1-upload-to	–	–	Deprecated	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
1fichier-com	–	ddl, encrypted	Sharehosting	–
4fastfile-com	–	–	Sharehosting	–
10upload-com	–	–	Sharehosting	–
1clickshare-net	–	–	Sharehosting	–
filedude-com	–	–	Sharehosting	–
asixfiles-com	–	–	Sharehosting	–
banashare-com	–	–	Sharehosting	–
biggerupload-com	–	–	Sharehosting	–
bitshare-com	–	ddl, video	Sharehosting	–
cash-file-net	–	–	Deprecated	–
cramit-in	–	–	Sharehosting	–
czshare-com	–	ddl, apache	Sharehosting	–
data-loading-com	–	–	Sharehosting	–
dataport-cz	–	–	Sharehosting	–
datei-to	–	ddl, apache, video	Sharehosting	–
downupload-com	–	–	Sharehosting	–
easybytez-com	–	ddl, apache	Sharehosting	–
edisk-cz	–	–	Deprecated	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
enterupload-com	–	–	Sharehosting	–
euroshare-eu	–	ddl	Sharehosting	–
extabit-com	–	–	Sharehosting	–
fastfileshare-com-ar	–	–	Sharehosting	–
fik1-com	–	–	Sharehosting	–
file2upload-net	–	–	Sharehosting	–
fileape-com	–	–	Sharehosting	–
fileflyer-com	–	–	Deprecated	–
filefrog-com	–	–	Deprecated	–
filehook-com	–	–	Sharehosting	–
fileover-net	–	–	Sharehosting	–
filepost-com	–	–	Sharehosting	–
files-mail-ru	–	ddl	Sharehosting	–
fileshare-in-ua	–	–	Sharehosting	–
filesmonster-com	–	ddl, encrypted	Sharehosting	–
filevo-com	–	–	Sharehosting	–
filezup-com	–	–	Deprecated	–
freakshare-net	–	ddl	Sharehosting	–
gaiafile-com	–	–	Sharehosting	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
gigapeta-com	–	ddl, video	Sharehosting	–
gotoupload-com	–	–	Sharehosting	–
hellshare-com	–	–	Sharehosting	–
hitfile-net	–	ddl, apache, video	Sharehosting	–
hulkshare-com	–	–	Sharehosting	–
hyperfileshare-com	–	–	Sharehosting	–
i-filez-com	–	–	Sharehosting	–
ifile-it	–	–	Sharehosting	–
kewlshare-com	–	–	Sharehosting	–
letitbit-net	–	ddl, apache, video	Sharehosting	–
loadfiles-in	–	–	Sharehosting	–
megafree-kz	–	–	Sharehosting	–
megarapid-eu	–	–	Deprecated	–
megashare-vnn-vn	–	–	Sharehosting	–
midupload-com	–	–	Sharehosting	–
multishare-cz	–	ddl, encrypted	Sharehosting	–
netuploaded-com	–	–	Sharehosting	–
oron-com	–	–	Sharehosting	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
przeklej-pl	–	–	Deprecated	–
putshare-com	–	–	Sharehosting	–
qshare-com	–	–	Sharehosting	–
quickshare-cz	–	–	Sharehosting	–
rapidshark-pl	–	–	Sharehosting	–
scribd-com	–	encrypted, ddl	Sharehosting	The application scribd.com is using external content and cache provider to deliver their content. This might influence the application scribd.com detection-rate.
share-online-bz	–	–	Deprecated	–
share-rapid-cz	–	–	Sharehosting	–
storage-to	–	ddl	Sharehosting	–
terafiles-net	–	–	Sharehosting	–
tunabox-net	–	–	Sharehosting	–
turbobit-net	–	ddl, apache	Sharehosting	–
ugotfile-com	–	–	Sharehosting	–
uloz-to	–	ddl, encrypted	Sharehosting	–
unibytes-com	–	–	Sharehosting	–
unlimit-co-il	–	–	Sharehosting	–
up-4share-vn	–	ddl	Sharehosting	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
upload-com-ua	–	ddl, apache	Sharehosting	–
uploadbox-com	–	ddl, apache	Sharehosting	–
uploader-pl	–	–	Sharehosting	–
uploadfloor-com	–	–	Sharehosting	–
uploadstation-com	–	–	Sharehosting	–
upnito-sk	–	–	Sharehosting	–
usershare-net	–	–	Sharehosting	–
videobb-com	–	ddl	Sharehosting	–
wupload-com	–	ddl, video	Sharehosting	–
x7-to	–	ddl, apache, video	Sharehosting	–
yunfile-to	–	–	Deprecated	–
zomgupload-to	–	–	Deprecated	–
ziddu-com	–	ddl, apache, video	Sharehosting	–
1-upload-com	–	–	Sharehosting	–
1-clickshare-com	–	–	Deprecated	–
123upload-pl	–	–	Deprecated	–
139pan-com	–	–	Sharehosting	–
163pan-com	–	–	Sharehosting	–
2shared-com	–	ddl, apache	Sharehosting	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
4share-ws	–	–	Sharehosting	–
4us-to	–	–	Sharehosting	–
alldrives-ge	–	–	Sharehosting	–
allshares-ge	–	–	Sharehosting	–
annonhost-net	–	–	Sharehosting	–
boosterking-com	–	–	Sharehosting	–
cobrashare-sk	–	–	Sharehosting	–
coolshare-cz	–	–	Sharehosting	–
crocko-com	–	ddl	Sharehosting	–
cum-com	–	–	Deprecated	–
eyvx-com	–	–	Sharehosting	–
filearn-com	–	–	Sharehosting	–
filejungle-com	–	–	Sharehosting	–
filemaze-ws	–	–	Sharehosting	–
filesflash-com	–	ddl	Sharehosting	–
freespace-by	–	ddl	Sharehosting	–
fsx-hu	–	–	Sharehosting	–
jakfile-com	–	–	Sharehosting	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
jumbofiles-com	–	–	Sharehosting	–
kickload-com	–	–	Sharehosting	–
missupload-com	–	–	Sharehosting	–
movreel-com	–	–	Sharehosting	–
platinshare-com	–	–	Sharehosting	–
sanupload-com	–	–	Sharehosting	–
shareflare-net	–	–	Sharehosting	–
u-115-com	–	–	Sharehosting	–
uploadc-com	–	–	Sharehosting	–
uploadking-com	–	–	Sharehosting	–
uptobox-com	–	encrypted, ddl	Sharehosting	–
filestube-com	–	ddl, apache	Sharehosting	–
gamefront-com	–	ddl, video	Sharehosting	–
megaporn-com	–	ddl, apache, video	Sharehosting	–
filesonic-in	–	ddl	Sharehosting	–
turbobit-ru	–	ddl, google	Sharehosting	–
Usenet	–	encrypted	Messaging	–
FTP	–	–	Deprecated	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
Filetopia	–	apache, encrypted	PeertoPeer	–
Manolito	–	–	PeertoPeer	–
iMesh	–	encrypted	PeertoPeer	–
Pando	–	apache, video, google	PeertoPeer	–
TVUPlayer	–	encrypted, apache, video	Streaming	–
PPStream	–	–	Deprecated	–
PPLive	–	–	Deprecated	–
Feidian	–	apache	Streaming	–
PoPo	–	apache	Messaging	–
Thunder/Webth under	–	encrypted, apache, audio, video, file-transfer	PeertoPeer	Traffic for p2p protocols such as Bittorrent or Edonkey should be detected as the respective protocol instead of thunder. On Windows some of the HTTP traffic may not be detected as Thunder.
UUSee	–	apache, video	Streaming	UUSEE uses the streaming service of Youku, letv and sohu. Also, pictures of the website are hosted by QQ and baidu. This applications can be detected within UUSEE traffic, but create more unrelated traffic.
OFF	–	–	PeertoPeer	–
HamachiVPN	–	encrypted	Tunnel	–
Citrix	–	–	Business	–
Quake	–	–	Gaming	–
SecondLife	–	encrypted, amazon-cloud, apache, google	Gaming	–
Battlefield	Battlefield™ Companion (Android) Battlefield™ Companion (iOS)	encrypted	Gaming	This protocol contains patterns from BattleLog, a social media platform for all battlefield games. BattleLog also contains a lot of YouTube and EA Origin traffic

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
PCAnywhere	–	–	RemoteControl	–
Icecast	–	audio, apache, video	Streaming	Shoutcast.com can contain Shoutcast and Icecast audio streams.Same for Icecast.org.
SHOUTcast	–	audio, apache, encrypted, video, google	Streaming	Shoutcast.com can contain SHOUTcast and Icecast audio streams. The same is true for Icecast.org. The video attribute detection for SHOUTcast is based on HTTP and does not work for TLS traffic.
QQGame	–	apache, file-transfer	Gaming	–
VeohTV	–	video, apache	Streaming	–
OpenFT	–	audio, video	PeertoPeer	–
IPP	–	–	NetworkManagement	–
Warcraft3	–	–	Gaming	–
ooVoo	–	–	Deprecated	–
Iskoot	–	encrypted, apache	VoiceoverIP	SSL Session ID Tracking (SIT) is required for the detection to work properly.
Fring	–	video, audio, apache, google	Depricated	–
IPTV	–		Streaming	–
YourFreedomTunnel	Your Freedom VPN Client (Android)	apache, encrypted	Tunnel	It is not possible to connect to YourFreedom through IPv6 FTP since there are no FTP IPv6 servers available.
VoipSwitchVoIP Tunnel	–	audio	VoiceoverIP	–
Meebo	–	audio, video, encrypted, apache, amazon-cloud, google	Deprecated	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
Aimini	–	apache, audio, video	PeertoPeer	–
ClubPenguin	–	apache, audio	Gaming	–
Florensia	–	–	Gaming	–
Octoshape	–	–	Streaming	–
PS3	–	–	Deprecated	–
Fiesta	–	–	Gaming	–
SplashFighter	–	–	Gaming	–
Activesync	–	–	Business	–
WAP-WSP	–	–	Deprecated	–
Guildwars	–	–	Gaming	–
rFactor	–	–	Gaming	–
TeamViewer	TeamViewer for Remote Control (Android) TeamViewer: Remote Control (iOS)	encrypted, file-transfer	RemoteControl	In remote control mode, teamviewer might send greater amounts of data and will then be detected as file-transfer. File-transfer can not be blocked completely. Small amounts of data can pass, but larger ones will be slowed significantly. Last improved in dpisig_2019-08-30.
GameKit	–	encrypted, apache	Gaming	–
UltraBac	–	encrypted	Business	–
ANtsP2P	–	encrypted	PeertoPeer	–
Scydo	–	encrypted	Deprecated	–
Mojo	–	file-transfer, apache, encrypted	PeertoPeer	–
CitrixGoTo	–	encrypted, apache	Conference	–
FiCall	–	–	VoiceoverIP	–
ComodoUnite	–	encrypted	Tunnel	–
Ventrilo	–	–	VoiceoverIP	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
MyPeople	–	–	Deprecated	–
JAP	–	–	Tunnel	–
Lync	–	–	Deprecated	–
Mig33	–	–	Deprecated	–
HTTPTunnel	–	–	Tunnel	–
Rhapsody	–	encrypted, apache, video	Streaming	–
WebQQ	–	encrypted, audio, apache	Deprecated	–
PDProxy	–	–	Tunnel	–
OracleDatabase	–	–	Database	–
RSVP	–	–	Deprecated	–
LotusNotes	–	apache	Conference	–
PoisonIvy	–	–	RemoteControl	–
MyVideo	–	–	Deprecated	–
OSupdate	–	android, ios, windows-mobile, microsoft-windows, encrypted, apache, google, chrome	NetworkManagement	The VPN application Hotspot Shield mimics patterns of Microsoft Windows updates. To avoid false positives, the detection of Microsoft Windows updates does not work in all cases for asymmetric traffic. The update for Microsoft OS can be detected and blocked, but downloaded data will be classified as microsoft-services.
DiDi	–	chat, audio, encrypted, apache, video	VoiceoverIP	A change of the subprotocol emphgeneric to im or voice for one flow is possible.
Spreecast	–	–	Deprecated	–
Simfy	–	encrypted, amazon-cloud, apache, audio	Streaming	Browsing the website will be detected as Simfy, which also includes streaming content.

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
Dota2	–	apache, google, video, stream, encrypted	Gaming	Last improved in dpisig_2019-08-02.
QVOD	–	encrypted, apache	Streaming	While browsing through the website of QVOD, a lot of traffic may be detected as unknown, because an external search service is used.
MicrosoftExchange	–	outlook_web_access, encrypted, ntlm	Business	SSL Session ID Tracking (SIT) is required for the detection to work properly.
NTLM	–	–	Deprecated	–
MeiLu	–	apache, amazon-cloud	Deprecated	–
NationalBaseball	–	apache	Deprecated	–
SPTGTV	–	apache	Streaming	–
CiaomobileHDTV	–	video, apache	Streaming	–
Skyking	–	video, apache	Streaming	Videos can be detected as YouTube, because Skyking uses streams of YouTube. Additionally, the Android application uses the service of Sopcast. This traffic will be detect as Sopcast.
Path	–	encrypted, amazon-cloud, video	Deprecated	–
iWowsystex	iWow (iOS) iWow (Android)	encrypted	Business	–
9ku	–	audio, video	Streaming	–
NextTV	Next TV (Android)	video, apache, audio	Streaming	–
RealMa	–	audio, apache	Streaming	–
DuoDuo	Dudu Txt Reader (iOS)	apache	Web	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
WealthGod	–	encrypted	Business	–
ThreeBamboo	–	–	Business	–
DisneyStorytime	–	encrypted, amazon-cloud	Deprecated	–
Rally	–	encrypted, apache	Business	–
MagineTV	–	amazon-cloud, encrypted, video, audio	Deprecated	
LivePerson	–	encrypted	Business	–
iOSAppStore	–	ios, encrypted, apache	Business	Last improved in dpisig_2019-08-30.
KingGaming	Candy Crush Saga (Android) Candy Crush Soda Saga (Android) Farm Heroes Saga (Android) Bubble Witch Saga (Android) Bubble Witch Saga (iOS) Candy Crush Saga (iOS) Farm Heroes Saga (iOS) Candy Crush Soda Saga (iOS)	encrypted, apache	Gaming	CDN Caching
WindowsStore	–	encrypted	Business	Last improved in dpisig_2019-10-18.
Meerkat	–		Deprecated	–
RovioGaming	–	encrypted	Gaming	CDN Caching
GitHub	–	encrypted	Web	–
GuildWars2	–	encrypted	Gaming	–
TrendMicroWorry-Free	–	apache, encrypted	Business	–
CrashPlan	–	encrypted	Deprecated	–
EAOrigin	–	encrypted, apache	Gaming	–
Tibia	–	apache, encrypted	Gaming	CDN Caching

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
QIP-RU	–	encrypted, audio	Web	–
musical_ly	–		Deprecated	–
EAGames	Command & Conquer™: Rivals PVP (iOS) The Sims FreePlay (iOS) The Sims Mobile (iOS) Command & Conquer: Rivals™ PVP (Android) The Sims FreePlay (Android) The Sims Mobile (Android)	encrypted, apache, video, audio	Gaming	The Sims FreePlay: When the application is launched, vungle, unity, popcap, google apis, google ads, crashlytics, google play, and akamai cloud flows are detected. These flows are not correlated to the EA Games. Purchase option is detected as google-services, google play, googlecloud. These flows are not correlated to the EA games. The Sims Mobile: This app is supported using three signatures - ea games, ea-origin, and electronic arts. Ads traffic (Vungle, unity etc and youtube ads) are not correlated to Sims Mobile. In App Purchase, payment gateway flows are not detected as Sims Mobile, rather as google play. These flows are not correlated. Traffic correlation issues are being tracked with AN-46767.
PopCap	–	encrypted, audio, apache	Gaming	–
ChatCube	–	audio, apache	Deprecated	–
PathofExile	–	encrypted	Gaming	–
NianticLabs	–	encrypted	Gaming	CDN Caching
2channel	–	apache	Web	2channel contains a lot of advertisement and external traffic which influences the detection rate.
ClashRoyal	–	–	Deprecated	
Upptalk	–	video, audio, encrypted, google, apache, amazon-cloud	Messaging	Upptalk is using iCloud hosts and also there is a lot of advertisement-traffic caused by this app. This might influence the Upptalk detection-rate.

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
share-online	–	ddl, video	Sharehosting	–
Ultrashare	–	ddl, apache, video	Sharehosting	The application Ultrashare is using some google shared services. Therefore most of the traffic caused by google services. This might influence the application Ultrashare detection-rate.
123upload	–	ddl	Sharehosting	–
iO	–	audio, encrypted, video	Deprecated	–
AIM	–	encrypted, file-transfer, apache, audio, google, video, amazon-cloud	Deprecated	
VPN-X	–	encrypted	Tunnel	–
NintendoNetwork	–	encrypted, video, apache, amazon-cloud, stream	Gaming	The Nintendo Anime-Channel is a free Video-on-Demand service that is available for Nintendo 3DS and Wii U. It allows streaming of different series. This video- content is not clearly related to Nintendo Networks and will not be detected as such.
PPFilm	–	video, stream, apache, encrypted	Streaming	–
AppleServices	–	encrypted, apache, ios, google, video, audio, blackberry, amazon-cloud	Business	The application attribute google will be set, if an Apple Operating System requests information from Google. For example verifying a TLS certificate. Last improved in dpisig_2019-08-30.
MicrosoftServices	–	encrypted, microsoft-windows, chat, web, apache, video, stream, audio, ntlm	Business	Last improved in dpisig_2019-10-18.

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
GoogleFCM	–	encrypted, google	Messaging	–
APNS	–	encrypted	Messaging	–
rusfolder	–	ddl, apache, video	Sharehosting	–
AkamaiCloud	–	encrypted, apache, audio, video, google, file-transfer	Business	Last improved in dpisig_2019-10-11.
LimelightCloud	–	encrypted, apache, audio, video	Business	Last improved in dpisig_2019-10-11.
Fastly	–	encrypted, video	Business	–
SAP	–		Business	–
WorldofKungFu	–		Gaming	–
Vevo	–	encrypted, video, stream, audio	Depricated	–
AppleGeolocation	–	encrypted, apache	Business	–
uploaded	–	ddl, apache, video	Sharehosting	–
easypaste-org	–	encrypted, ddl, apache	Sharehosting	–
GoogleAPIs	–	encrypted, google	Business	–
StackPathCloud	–	encrypted, video	Business	Last improved in dpisig_2019-08-30.
AmazonServices	–	encrypted, audio, stream, apache	Web	–
Crashlytics	–	encrypted	Advertisement and Analytic Services	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
FacebookCloud	–	encrypted, apache, file-transfer, video, audio	Business	Facebook cloud is a CDN and cloud service providing online storage and content delivery. This classification contains fbcdn and akamai traffic and is mostly used for static content like photos or videos. Embedded video content may be classified as filetransfer.
Turner	–	encrypted, apache, video	Streaming	the detection rate is low, when a dedicated service like cnn, a sub-division of turner, can be detected instead. Some traffic is covered by amazon cloud.
LoopTradeClassifieds	–	–	E-Commerce	–
FunPlus	–	encrypted	Mobile	–
Cedexis	–	encrypted	Business	–
Symantec	–	encrypted, apache	Business	–
GoogleAppEngine	–	encrypted, google	Web	Parts of the Google App Engine traffic is detected as Google Shared Services with the attribute encrypted. Therefore, it is not always possible to block Google App Engine.
EpicBrowserProxy	–	encrypted	Tunnel	–
PPTV	–	file-transfer, video, apache, google, encrypted	Streaming	PPTV uses the streaming service of Youku. Thus, the application Youku can be detected within PPTV traffic. Browsing the PPTV website is classified as PPTV too.
AvastServices	–	encrypted, apache	Business	–
EasyVPN	–		Deprecated	–
HolaServices	–	encrypted, google	Business	–
CDNvideo	–	encrypted, video, apache	Web	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
Vungle	–	encrypted, video	Advertisement and Analytic Services	–
Vivox	–	encrypted, audio	VoiceoverIP	–
EnmasseGames	–	encrypted, amazon-cloud, apache	Gaming	–
ApexLegends	–		Gaming	Logging is done via generic EA services. Thus, game logging can not be blocked.
CounterStrikeGO	–	apache	Gaming	In a blocking scenario, it is possible to log in and navigate through the main menu.
VPNTorBrowser	VPN + TOR Browser Private Web (iOS)	encrypted	Tunnel	The application VPN Tor Browser uses Actmobile Services as underlying protocol. Especially in block cases, the application can also be classified as Actmobile Services.
Office365	–	encrypted	Business	–
RedTube	–	video, encrypted, apache	Streaming	The video attribute classification for encrypted traffic is based on throughput calculation. Throttling or other throughput influencing methods may result in preventing classification of video attribute. Last improved in dpisig_2019-10-11.
Vine	Vine (Android)	video, encrypted	Social	–
Net2Phone	net2phone (iOS) net2phone (Android)	audio	VoiceoverIP	–
iLive-to	–	–	Deprecated	Deprecated
vBulletin	vBulletin Community Forum (Android) vBulletin Community Forum (iOS)	apache, video, google	Social	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
Redmine	RedminePM (iOS) RedminePM - Redmine client app (Android)	apache	Business	–
KasperskyAntivirus	Kaspersky Mobile Antivirus: AppLock & Web Security (Android)	–	Business	–
SoundCloud	SoundCloud - Music & Audio (IOS) SoundCloud - Music & Audio (Android)	encrypted, audio	Streaming	–
RunKeeper	Runkeeper—GPS Running Tracker (IOS) Runkeeper - GPS Track Run Walk (Android)	encrypted, amazon-cloud, apache, google, chat	Social	To ensure full detection results the SSL Session ID Tracker (SIT) has to be enabled. The application contains a lots of analytics and ad traffic which is detected separately as HTTP or SSL.
myBook	myBook (iOS) myBook (Android)	encrypted, apache	Web	–
NikePlus	Nike Run Club (iOS) Nike Run Club (Android)	encrypted, apache, amazon-cloud	Mobile	–
TVBS	TVBS (iOS) TVBS (Android)	video, apache, encrypted, amazon-cloud	Streaming	–
eDonkey	–	encrypted, apache	PeertoPeer	–
Slingbox	SlingPlayer Free for Phone (Android) Slingplayer Basic for iPhone (iOS)	–	Streaming	–
fileho-com	–	–	Deprecated	Deprecated
SoftEthernet	–	encrypted	Tunnel	–
Yandex	Yandex Browser (iOS) Yandex Browser With Protect (Android)	encrypted, apache, audio	Web	–

Application	Client/App Name & App Store Link	Attributes	Group/Tag	Limitations/Remarks
360TotalSecurity	360 Security (Android)	encrypted, apache	Business	For Android and Mac OSX, a blocking of the main application functionality is not possible, because the network traffic of 360 total security only consist of signature updates and login mechanisms. For the same reason the actual detected traffic will not be high.
Unity	–	encrypted, amazon-cloud apache, video	Gaming	Last improved in dpisig_2019-08-30.
Wargaming-net	World Of Tanks Assistant (Android) Gods and Glory: War for the Throne (Android) World of Warships Blitz (Android) World of Tanks Blitz MMO (Android) World of Tanks Assistant (iOS) Gods and Glory: War of Thrones (iOS) World of Warships Blitz: MMO (iOS) World of Tanks Blitz MMO (iOS)	encrypted, apache	Gaming	–
NovatechCore	LinkVPN - Fast & Unlimited VPN Proxy (iOS) LinkVPN Free VPN Proxy (Android) FlashVPN Free VPN Proxy (Android)	encrypted	Tunnel	–

Protocols

This section provides the complete list of Protocols supported by the DPI Signature.

Protocol	Group	Attributes	Limitations/Remarks
BitTorrent	PeertoPeer	tcp_wo_hs, uTP, socks_v5, http-connect	Some bittorrent clients use Teredo tunnels for their file transfers. To properly detect these transfers it is necessary to enable the plain tunnel decapsulation
eDonkey	PeertoPeer	tcp_wo_hs, socks_v5	–
Gnutella	PeertoPeer	tcp_wo_hs	–
WinMX	PeertoPeer	–	–
DirectConnect	PeertoPeer	adc-peer, hub, peer	–
AppleJuice	PeertoPeer	tcp_wo_hs	–
Soulseek	PeertoPeer	tcp_wo_hs	–
Ares	PeertoPeer	tcp_wo_hs	Asymmetric detection is supported only for traffic from the client side.
Freenet	PeertoPeer	–	–
Mute	PeertoPeer	–	–
XDCC	Messaging	–	–
Skype	VoiceoverIP	tcp_wo_hs, socks_v5, http-connect	Skype connections in general are not exclusively used for a single type of connections. That means that video and voice can be transferred in the same or separate flows. So if a video flow is detected, the same flow may also contain voice packets. Also, file transfers can be using separate flows or can be multiplexed in an existing voice call. The Skype client is also capable of instant messaging with a Facebook account. This chat can't get safely detected as Skype without causing false positives, because it uses the Facebook API. PACE classifies the traffic as protocol XMPP with Application Facebook. Voice and video calls are classified after some initial time and the type will be reevaluated recurrently. That means, that a flow marked as sub protocol unknown can be marked as voice later and switch again if video is detected because it is possible to enable video transmission at any given time during a phone call. Traffic to gateway.messenger.live.com is detected as Skype due to the integration of MSN into the Skype client. Skype and Skype for Business share the same technology and as such their traffic can not always be distinguished. When possible both protocols should be treated as one to avoid issues arising from this.
SIP	VoiceoverIP	tcp_wo_hs	–

Protocol	Group	Attributes	Limitations/Remarks
MSN	Messaging	tcp_wo_hs	–
Yahoo	Messaging	tcp_wo_hs	The Hotmail webclient has the capability to chat via MSN. Blocking MSN also disables the "manage" and "preferences" functionality of Hotmail. In addition it is possible to post on a Facebook wall via the Hotmail account using the Facebook api. This is not detected because the post is send from within a running encrypted SSL flow.
Oscar	Messaging	tcp_wo_hs, http-connect	–
IRC	Messaging	tcp_wo_hs, socks_v5	–
XMPP	Messaging	tcp_wo_hs, http-connect	In unidirectional file-transfer traffic only the direction including the file-transfer data is classified as XMPP with the protocol attribute file-transfer.
HTTP	Web	tcp_wo_hs, media, socks_v4, socks_v5, http-connect, tunnel	–
POP	Mail	tcp_wo_hs, socks_v4, socks_v5	–
SMTP	Mail	tcp_wo_hs, socks_v5, socks_v4	–
IMAP	Mail	tcp_wo_hs, socks_v5	–
Usenet	Deprecated	–	–
FTP	Filetransfer	ftp_control, tcp_wo_hs, ftp_data, socks_v5	–
DNS	NetworkManagement	tcp_wo_hs, tunnel	–
RTSP	Streaming	tcp_wo_hs	–
Filetopia	PeertoPeer	–	–
Manolito	PeertoPeer	–	–
iMesh	PeertoPeer	tcp_wo_hs	–
SSL	Encrypted	tcp_wo_hs, spdy, http2, socks_v5, socks_v4, http-connect, tls_v1_3, esni, http, stun, webrtc, ftp, imap, pop3, managesieve, coap, xmpp	SSL/TLS is a network protocol used to provide secure communication over IP networks. Since PACE 1.49 SSL may be classified for flows in which the handshake is missing. In this case no certificate will be present it will not be possible to extract meta-data from such flows. The SSL midstream classification is capable to classify TLS >= 1.0 and SSL 3.0 midstream flows. Last improved in dpisig_2019-08-30.

Protocol	Group	Attributes	Limitations/Remarks
Flash	Streaming	tcp_wo_hs, media, socks_v5, socks_v4, http-connect	–
MMS	Streaming	–	–
MPEG	Streaming	tcp_wo_hs, media, socks_v5	–
QuickTime	Streaming	media, tcp_wo_hs	–
Pando	PeertoPeer	tcp_wo_hs	–
OpenVPN	Tunnel	tcp_wo_hs, http-connect	–
IPsec	Tunnel	ipsec_payload, tcp_wo_hs	–
GRE	Tunnel	–	–
undetected	Generic	tcp_wo_hs, socks_v5, socks_v4, http-connect	–
H323	VoiceoverIP	tcp_wo_hs, ras	Added <i>ras</i> attribute in dpisig_2019-07-19
WindowsMedia	Streaming	tcp_wo_hs, media	–
RealMedia	Streaming	tcp_wo_hs	–
TvAnts	Streaming	–	–
SopCast	Streaming	–	–
TVUPlayer	Streaming	–	–
PPStream	Deprecated	–	–
PPLive	Streaming	–	–
IAX	VoiceoverIP	–	–
VTun	Tunnel	–	–
MGCP	VoiceoverIP	–	–
Gadu-Gadu	Messaging	–	Only file transfers with a minimum throughput of 50 kB/s and a duration longer than 1 second can be detected
Zattoo	Streaming	tcp_wo_hs	–
WinNY	PeertoPeer	–	–
QQ	Messaging	tcp_wo_hs	–
QQLive	Streaming	–	The goal of the QQLive detection is to detect the video traffic. The website should be accessible during a block test

Protocol	Group	Attributes	Limitations/Remarks
Feidian	Streaming	–	–
SSH	RemoteControl	tcp_wo_hs, http-connect	Last improved in dpisig_2019-10-11.
PoPo	Messaging	–	–
Thunder/Webth under	PeertoPeer	tcp_wo_hs	Traffic for p2p protocols such as Bittorrent or Edonkey should be detected as the respective protocol instead of thunder. On Windows some of the HTTP traffic may not be detected as Thunder.
UUSee	Streaming	–	–
VNC	RemoteControl	–	–
ICMP	NetworkManagement	tunnel	–
DHCP	NetworkManagement	–	–
IGMP	NetworkManagement	–	–
EGP	NetworkManagement	–	–
Steam	Gaming	–	Encrypted traffic from the steam website will also be detected as protocol Steam, because the applications use the website for community and shopping activities.
Half-Life2	Deprecated	–	–
XBOX	Gaming	–	–
SoftEthernet	Tunnel	–	–
SCTP	NetworkManagement	–	–
SMB/CIFS	Filetransfer	tcp_wo_hs	–
Telnet	RemoteControl	tcp_wo_hs, tn3270	–
NTP	NetworkManagement	–	–
NFS	Filetransfer	tcp_wo_hs	–
RTP	Streaming	http-connect, tcp_wo_hs, webrtc	Last improved in dpisig_2019-10-11.

Protocol	Group	Attributes	Limitations/Remarks
Tor	Tunnel	–	Testing may fail for obfs2 and obfs3 traffic when packets are out of order, verify the result by running CAAP or PAT on the trace file with PARO enabled. obfs4 and other obfuscation proxies are currently not detected as TOR and will be classified as unknown
SSDP	NetworkManagement	–	–
WorldofWarcraft	Gaming	–	–
AVI	Streaming	media	–
VPN-X	Tunnel	–	–
HamachiVPN	Tunnel	tcp_wo_hs	–
Skinny	VoiceoverIP	tcp_wo_hs	–
Citrix	Deprecated	–	–
OGG	Streaming	media, tcp_wo_hs	–
Paltalk	Messaging	tcp_wo_hs	Last improved in dpisig_2019-08-30.
Move	Streaming	–	–
PostgreSQL	Database	–	–
OSPF	NetworkManagement	–	–
BGP	NetworkManagement	tcp_wo_hs	–
Quake	Gaming	–	–
SecondLife	Gaming	–	–
MySQL	Database	tcp_wo_hs	–
Battlefield	Gaming	–	–
PCAnywhere	RemoteControl	–	–
RDP	RemoteControl	tcp_wo_hs	–
SNMP	NetworkManagement	–	–
Kontiki	Streaming	–	–
VeohTV	Streaming	–	–
Kerberos	NetworkManagement	–	–

Protocol	Group	Attributes	Limitations/Remarks
IPinIP	Tunnel	–	–
GTP	Tunnel	gtp_cp, gtp_prime, gtp_up, gtp_v1, gtp_v2	–
RADIUS	NetworkManagement	–	–
STUN	VoiceoverIP	tcp_wo_hs, http-connect	–
Syslog	NetworkManagement	tcp_wo_hs	–
TDS	Database	–	–
NetBIOS	NetworkManagement	tcp_wo_hs	–
MulticastDNS	NetworkManagement	–	–
IPP	NetworkManagement	–	–
LDAP	NetworkManagement	Tcp_wo_hs	–
Warcraft3	Gaming	–	–
XDMCP	RemoteControl	tcp_wo_hs	–
ooVoo	Deprecated	–	–
Fring	VoiceoverIP	tcp_wo_hs	–
Mapi	Mail	–	–
L2TP	Tunnel	–	–
ISAKMP	Tunnel	–	–
RTCP	Streaming	–	–
TFTP	Filetransfer	–	–
MSSQL	Deprecated	–	–

Protocol	Group	Attributes	Limitations/Remarks
PPP	Tunnel	–	PACE will only detected PPP packets encapsulated in GRE(Generic Routing Encapsulation). Any other packets that may contain PPP will not be detected as protocol PPP. PPPoE traffic will be returned as unknown as it is a layer 2 protocol and PACE will only process the layer 3 portion of the packet. L2TP packets will either be decapsulated or returned as unknown if the decapsulation is disabled. The information that the packet is a GRE packet is not directly available but it can always be assumed to be a GRE packet when PPP is detected.
PPTP	Tunnel	–	–
StealthNet	PeertoPeer	tcp_wo_hs	–
SAP	Business	tcp_wo_hs	–
ICMPv6	NetworkManagement	–	–
YourFreedomTunnel	Tunnel	ftp_control, http-connect	It is not possible to connect to YourFreedom through IPv6 FTP since there are no FTP IPv6 servers available.
VoipSwitchVoIPTunnel	VoiceoverIP	–	–
DHCPv6	NetworkManagement	–	–
Meebo	Messaging	–	–
Funshion	Streaming	tcp_wo_hs	Funshion uses the streaming service of Youku. Thus, the application Youku can be detected within Funshion traffic.
AFP	Filetransfer	tcp_wo_hs	–
Aimini	PeertoPeer	–	–
ClubPenguin	Gaming	–	–
Florensia	Gaming	–	–
TeamSpeak	VoiceoverIP	–	–
MapleStory	Gaming	–	–
Octoshape	Streaming	–	–
PS3	Deprecated		–
Dofus	Gaming	–	–
WII	Deprecated		–
WorldofKungFu	Gaming	–	–

Protocol	Group	Attributes	Limitations/Remarks
Fiesta	Gaming	–	–
SplashFighter	Gaming	–	–
CrossFire	Gaming	–	–
WAP–WSP	Mobile	–	–
WAP–WTP–WSP	Mobile	–	–
WAP–WTLS	Mobile	–	–
MultimediaMessaging	Mobile	tcp_wo_hs	–
Guildwars	Gaming	–	–
Armagetron	Gaming	–	–
RIMproprietary	Mobile	–	–
rFactor	Gaming	–	–
RealDataTransport	Streaming	–	–
TeamViewer	RemoteControl	tcp_wo_hs, http-connect	–
GameKit	Gaming	–	–
NetMotion	Tunnel	–	–
Teredo	Tunnel	–	–
Spotify	Streaming	tcp_wo_hs, http-connect	–
IMplus	Messaging	tcp_wo_hs	–
UltraSurf	Tunnel	–	Running the Windows client of Ultrasurf without browsing can not be safely classified and blocked, as Ultrasurf uses 3rd party servers for connection establishment
Viber	VoiceoverIP	tcp_wo_hs	If only the protocol Viber is blocked, the application Viber starts an SSL connection and is therefore still running. It can be blocked by blocking the application Viber. Limitations for blocking the application Viber with specific application attributes apply for the protocol as well, as the application uses the protocol Viber.
Tango	VoiceoverIP	tcp_wo_hs	Last improved in dpisig_2019-10-11.
iPlayer	Streaming	–	–
JBK3000	Industrial	–	–

Protocol	Group	Attributes	Limitations/Remarks
OperaMini	Mobile	tcp_wo_hs	–
Bolt	Mobile	tcp_wo_hs	–
Scydo	VoiceoverIP	tcp_wo_hs	–
WhatsApp	Messaging	tcp_wo_hs	Apple mobile devices are capable of receiving WhatsApp messages even when the App is not started. This is not WhatsApp specific traffic. Apple uses iMessage to relay the messages. In blocking scenario of file-transfer, after sending or receiving a file, also chat messages will be blocked for the current conversation
Webex	Conference	–	–
Mojo	PeertoPeer	–	–
IMO	Messaging	tcp_wo_hs	–
CitrixGoTo	Conference	–	Accesses to citrixonline.com and citrixonlinecdn.com can not be referred to the application GoToMeeting clearly. If a correlation is not possible for these accesses, the affected flows will be detected as protocol CitrixGoTo without a correlated application
ComodoUnite	Tunnel	–	–
Goober	Deprecated	–	–
Ventrilo	VoiceoverIP	–	–
MyPeople	Deprecated	–	–
WebSocket	Web	tcp_wo_hs	–
eBuddy	Messaging	tcp_wo_hs	–
SSTP	Tunnel	–	–
AdobeConnect	Business	–	–
JAP	Tunnel	–	–
LDP	NetworkManagement	tcp_wo_hs	–
Wuala	Deprecated	–	Deprecated in dpisig_2019-08-02
MSRP	Messaging	–	–
Lync	Deprecated	–	–
Socks	Tunnel	tcp_wo_hs, socks_v4, socks_v5	–
Audiogalaxy	Streaming	–	–

Protocol	Group	Attributes	Limitations/Remarks
Mig33	Deprecated	–	–
Rhapsody	Streaming	–	–
Sudaphone	Deprecated	–	–
WebQQ	Deprecated	–	Deprecated in dpisig_2019-08-02
PDProxy	Tunnel	–	–
Silverlight	Streaming	tcp_wo_hs	–
CyberGhost	Deprecated	–	–
FIX	Business	tcp_wo_hs	–
OracleDatabase	Database	tcp_wo_hs	–
RSVP	NetworkManagement	–	–
LotusNotes	Conference	–	–
PoisonIvy	RemoteControl	–	–
NetFlow	NetworkManagement	–	–
WeChat	Messaging	tcp_wo_hs	To drop the hold and talk feature the subprotocols generic and data_message should be blocked at the same time. Last improved in dpisig_2019-10-11.
SOAP	RemoteControl	tcp_wo_hs	–
KikMessenger	Messaging	–	–
MPlus	Messaging	tcp_wo_hs	–
Line	VoiceoverIP	tcp_wo_hs	–
Iperf	NetworkManagement	tcp_wo_hs	UDP packets with less than 16 byte layer 7 payload are not detected as Iperf and remain as unknown traffic. If the hostname is longer than 16 characters, TCP traffic can not be detected.
Diameter	NetworkManagement	tcp_wo_hs	–
HikeMessenger	Messaging	–	If the detection did not work and the packets are tagged as SSL the test should be run with the session ID tracker (SIT). For more information about SIT read the PACE integration manual.
MQTT	Industrial	tcp_wo_hs	–
DCERPC	RemoteControl	tcp_wo_hs, dfs-r	–

Protocol	Group	Attributes	Limitations/Remarks
IDrive	Filetransfer	–	SSL Session ID Tracking (SIT) is required for the detection to work properly. In a blocking scenario SIT is not necessary because the client won't be able to obtain a session ID.
Dota2	Gaming	–	–
QVOD	Streaming	tcp_wo_hs	When browsing on the website a lot of traffic may be detected as HTTP, because an external search service is being used.
MicrosoftExchange	Business	–	SSL Session ID Tracking (SIT) is required for the detection to work properly.
MicrosoftDynamicsNAV	Business	–	–
Tibia	Gaming	–	–
Kakao	Messaging	tcp_wo_hs	–
Destiny	Gaming	–	It is possible that game updates will be detected as HTTP because they are not directly related to the gameplay
VyprVPN	Tunnel	–	VyprVPN can use different security and tunneling protocols like L2TP, PPTP, OpenVPN or IPSEC. Additional there is a modified OpenVPN 256-Bit mode called Chameleon. Using VyprVPN with the different protocols: – Android – OpenVPN is detect as VyprVPN with real protocol OpenVPN – Chameleon is detect as real protocol VyprVPN – Mac OSX – OpenVPN is detect as VyprVPN with real protocol OpenVPN – Chameleon is detect as real protocol VyprVPN – L2TP as L2TP and IPSEC – PPTP as PPTP and PPP – Windows – OpenVPN is detect as VyprVPN with real protocol OpenVPN – Chameleon is detect as real protocol VyprVPN – L2TP as L2TP and IPSEC – PPTP as PPTP and PPP – iOS – Detect as protocol IPSEC For all platforms the generic VyprVPN traffic is detect as VyprVPN.
TACACS	NetworkManagement	–	–
Threema	Messaging	–	The location search through the app is detect as google or sometimes as unknown.
NTLM	NetworkManagement	–	–
Telegram	Deprecated	–	–
WHOIS	NetworkManagement	–	–
IPv4	Generic	–	–
IPv6	Generic	–	–

Protocol	Group	Attributes	Limitations/Remarks
TCP	Generic	tcp_wo_hs, socks_v4, socks_v5, http-connect	–
UDP	Generic	–	–
RTMP	Streaming	–	–
NNTP	Business	socks_v5	–
FLUTE	Filetransfer	–	–
Megaco	NetworkManagement	–	–
Twitch	Streaming	–	–
QUIC	Business	–	This classification is designed to cover the Google as well as the IETF version. Last improved in dpisig_2019-07-19.
FuzeMeeting	Business	–	–
AgeofWarringEmpire	Gaming	–	This application produces a lot of additional advertisement traffic. Therefore the basic detection rate is low in comparison to the caused traffic.
Doook	Messaging	–	–
Speedtest	NetworkManagement	–	–
IEC61850	Industrial	iec61850-mms, iec61850-goose, iec61850-smv	–
SCADA	Industrial	tcp_wo_hs	–
Shoutcast	Streaming	tcp_wo_hs	–
CassandraQueryLanguage	Database	–	–
CassandraInterNodeCommunication	Database	–	SSL encrypted internode communication is currently not supported by the detection.
Thrift	Generic	–	This protocol is designed to create small tcp packets. Therefore flows will not create huge traffic.
PlayStationNetwork	Gaming	–	–
Supercell	Gaming	tcp_wo_hs	–

Protocol	Group	Attributes	Limitations/Remarks
SkypeforBusiness	Business	–	Audio and video calls between Skype for Business and regular Skype are either classified as Skype or RTP streams. Due to both applications being migrated to the same technology it is not always possible to distinguish between Skype and Skype for Business traffic. When possible both protocols should be treated as one to avoid issues arising from this.
T3	NetworkManagement	–	–
SRTP	Streaming	–	–
SRTCP	Streaming	–	–
LeagueofLegends	Gaming	–	–
Facebook	Deprecated	–	–
BattleNet	Gaming	–	–
Wowza	Streaming	–	A known issue is that it is not possible to classify RTMP flow from Wowza Streaming Engine as Wowza
Zalo	Messaging	–	Sometimes it is not possible to block chat messages on Android devices
GuildWars2	Gaming	–	–
CrashPlan	Business	tcp_wo_hs	–
EAOrigin	Gaming	–	–
Fasttrack	PeertoPeer	–	–
Icecast	Streaming	–	–
ChatCube	VoiceoverIP	tcp_wo_hs	–
Good	Business	–	–
PathofExile	Gaming	–	–
PCoIP	RemoteControl	–	–
OCSP	Web	tcp_wo_hs, socks_v4, socks_v5, http-connect	–
Modbus	Industrial	tcp_wo_hs	–
FASP	Filetransfer	–	–
ICA	Business	–	–
OPCUA	Industrial	–	–

Protocol	Group	Attributes	Limitations/Remarks
OpenFlow	NetworkManagement	–	–
IPMI	NetworkManagement	–	–
AMQP	Industrial	–	–
S7comm	Industrial	–	S7comm will only be classified if the S7 payload is available on top of TPKT and COPT.
OFF	PeertoPeer	tcp_wo_hs	–
Rsync	Filetransfer	–	–
ZeroMQ	Generic	–	–
DNP3	Industrial	–	–
BACNET	Industrial	–	–
NintendoNetwork	Gaming	–	–
PPFilm	Streaming	–	–
Bitcoin	Business	tcp_wo_hs	–
LeTV	Streaming	–	–
Psiphon	Tunnel	http2	Psiphon uses a variation of obfuscation techniques and tunnel modes and is capable of switching between them. In addition, most of these variations need to be correlated for a safe detection. Therefore, blocking is limited. Safe classification of asymmetric or singular flows is also limited.
Zero	Business	tcp_wo_hs, http2	–
Synology	Business	–	–
CoAP	Industrial	–	Both traffic directions are needed for the CoAP classification
Lineage2	Gaming	–	Only game server traffic is classified as the login protocol causes too little traffic.
IPFIX	NetworkManagement	–	–
Minecraft	Gaming	tcp_wo_hs	–
StratumMining	Business	tcp_wo_hs	–
LocalPeerDiscovery	PeertoPeer	–	–
ZeroVPN	Tunnel	–	–

Protocol	Group	Attributes	Limitations/Remarks
TurboVPN	Tunnel	–	–
VPNMaster	Tunnel	–	Last improved in dpisig_2019-10-11.
Iqiyi	Streaming	tcp_wo_hs	–
SnapVPN	Tunnel	–	–
VPNUnlimited	Tunnel	–	–
ExpressVPN	Tunnel	–	–
TunnelGuru	Tunnel	–	–
BOTIM	VoiceoverIP	–	–
HiVPN	Tunnel	–	–
SpeedVPNMobileAppDev	Tunnel	–	–
DashVPN	Deprecated	tcp_wo_hs	–
YeeCall	VoiceoverIP	tcp_wo_hs	Last improved in dpisig_2019-08-30
AtomVPN	Tunnel	–	–
Fortnite	Gaming	–	–
WireGuard	Tunnel	–	–
AmazeVPN	Deprecated	–	–
FastSecureVPN	Tunnel	–	–
ZPN	Tunnel	–	–
WorldofTanks	Gaming	–	–
AppVPN	Tunnel	tcp_wo_hs	–
X-VPN	Tunnel	tcp_wo_hs	–
DroidVPN	Tunnel	tcp_wo_hs	–
DTLS	Encrypted	tcp_wo_hs	–
SmartVeePN	Tunnel	tcp_wo_hs	–
ApexLegends	Gaming	–	–
VPN365	Tunnel	–	–
CounterStrikeGO	Gaming	–	–

Protocol	Group	Attributes	Limitations/Remarks
SuperVPNbyCheng	Tunnel	tcp_wo_hs	–
ActVPN	Tunnel	–	–
HotspotVPN	Tunnel	tcp_wo_hs	–
MTPProto	Messaging	tcp_wo_hs	–
Shadowsocks	Tunnel	–	–
Asphalt9	Deprecated	tcp_wo_hs	–
ActmobileServices	Tunnel	tcp_wo_hs	–
ZRTP	Streaming	–	–
PUBG Mobile	Gaming	–	–
Gameloft Services	Gaming	tcp_wo_hs	Last improved in dpisig_2019-10-11.
Wickr	Voice over IP	encrypted, audio	–
VPNProxyMasterUnlimited	Tunnel	encrypted	–
Zoom	Conference	encrypted	It is not possible to distinguish between a conference and a video conference.
VPNProxyMaster	Deprecated	encrypted	–
DeadRivals	Deprecated	–	–
IGG	Gaming	tcp_wo_hs	Last improved in dpisig_2019-10-11.
Mudfish	Tunnel	tcp_wo_hs	–
SOMA	Messaging	tcp_wo_hs	Last improved in dpisig_2019-10-11.
HatchCloudGaming	Gaming	tcp_wo_hs	–
SourceEngine	Gaming	–	–

Protocol Attributes

Protocol Attributes are context-sensitive attributes which are only set when a specific protocol was detected on the protocol stack layer 7. For example, SPDY is only detected and set when SSL traffic is classified.

Table 3 **DPI Supported Protocol Attributes**

ID	Protocol Attribute Name	Description
0	none	Protocol has no attribute.
1	ipsec_payload	Indicates that the current connection is IPSEC payload traffic
2	ipsec_isakmp	Indicates that the current connection is IPSEC isakmp traffic.
3	ftp_control	Indicates that the current connection is FTP control traffic
4	ftp_data	Indicates that the current connection is FTP data traffic
5	gtp_up	Indicates that the current connection is GTP user plane traffic.
6	gtp_cp	Indicates that the current connection is GTP control plane traffic
7	gtp_prime	Indicates that the current connection is GTP prime traffic.
8	gtp_v1	Indicates that the current connection is GTP v1 traffic
9	gtp_v2	Indicates that the current connection is GTP v2 traffic.
10	socks_v4	Indicates that the current connection is SOCKS v4 traffic
11	socks_v5	Indicates that the current connection is SOCKS v5 traffic.
12	media	Protocol attribute media is deprecated and should not be used anymore
13	uTP	Indicates that the current connection is uTP traffic
14	hub	Indicates that the current connection is Direct Connect Hub traffic.
15	peer	Indicates that the current connection is Direct Connect peer traffic
16	adc-peer	Indicates that the current connection is Direct Connect Advanced direct connect peer traffic.
17	spdy	Indicates that the current connection is SPDY traffic
18	tn3270	Indicates that the current connection is Telnet TN3270 traffic
19	tcp_wo_hs	Indicates that the current connection is TCP traffic for which no handshake has been observed. It is set if no part of the TCP handshake has been seen, i.e. when the SYN-flag is missing in the TCP flow as well as the SYN/ACK-flag.

Table 3 DPI Supported Protocol Attributes

ID	Protocol Attribute Name	Description
20	iec61850-goose	Indicates that the current connection is IEC61850-8-1 Generic Object Oriented Substation Event traffic
21	iec61850-mms	Indicates that the current connection is IEC61850-8-1 Manufacturing Messaging Specification traffic.
22	iec61850-smv	Indicates that the current connection is IEC61850-7-2 Sampled Measured Values traffic
23	http2	Indicates that the current connection is HTTP2 traffic. The attribute can be set for the SSL and Zero layer 7 protocol.
24	http-connect	Indicates that the current connection is HTTP-Connect traffic.
25	tunnel	Indicates that the current connection is tunnel traffic.
26	tls_v1_3	Indicates that the current connection is TLS traffic with version 1.3.
27	dfs-r	Specifies the SD Microsoft Distributed File System Replication Protocol.
28	ras	Specifies the Registration, Admission and Status protocol for Endpoint-Gatekeeper and H.323 Gatekeeper communication.
29	esni	Indicates that the current connection contains an TLS client hello with encrypted SNI extension.
30	http	Indicates that the current connection contains an TLS/DTLS handshake with negotiated HTTP traffic via ALPN.
31	stun	Indicates that the current connection contains an TLS/DTLS handshake with negotiated STUN traffic via ALPN.
32	webrtc	Indicates that the current connection contains an TLS/DTLS handshake with negotiated WebRTC traffic via ALPN.
33	ftp	Indicates that the current connection contains an TLS/DTLS handshake with negotiated FTP traffic via ALPN.
34	imap	Indicates that the current connection contains an TLS/DTLS handshake with negotiated IMAP traffic via ALPN.
35	pop3	Indicates that the current connection contains an TLS/DTLS handshake with negotiated POP3 traffic via ALPN.
36	managesieve	Indicates that the current connection contains an TLS/DTLS handshake with negotiated managesieve traffic via ALPN.
37	coap	Indicates that the current connection contains an TLS/DTLS handshake with negotiated COAP traffic via ALPN.
38	xmpp	Indicates that the current connection contains an TLS/DTLS handshake with negotiated XMPP traffic via ALPN

Application Attributes

Application Attributes are context-free attributes that can occur independently of the application. For example, Application Attribute, “audio” will be set for Viber/Skype audio calls. Application Attribute, “encrypted” will be set to indicate that the current connection is encrypted traffic. For some flows, multiple attributes can be present at the same time. This allows for simultaneous attributes such as encrypted and audio.

Table 4 DPI Supported Application Attributes

ID	Application Attribute Name	Description
0	none	Application has no attribute.
1	encrypted	Indicates that the current connection is encrypted traffic.
2	audio	Indicates that the current connection is an audio or voice signal.
3	out	Indicates that the current connection is a landline call, e.g. a call to a home phone.
4	video	Indicates that the current connection is a video signal.
5	file-transfer	Indicates that the current connection is a file transfer
6	web	Indicates that the current connection is a surf the Internet session.
7	chat	Indicates that the current connection is a chat session
8	mail	Indicates that the current connection is mail traffic.
9	stream	Indicates that the current connection is a continues unidirectional stream of audio and / or video.
10	android	Indicates that the client side uses the operating system Android.
11	ios	Indicates that the client side uses the operating system iOS.
12	windows-mobile	Indicates that the client side uses the operating system Windows Mobile.
13	blackberry	Indicates that the client side uses the operating system Blackberry.
14	picture	Indicates that the current connection transfers pictures.
15	ddl	Indicates that the current connection is a Direct Download Hoster.
16	google	Indicates that the current connection is a Google service.
17	outlook_web_access	Indicates that the current connection uses the Microsoft Exchange Outlook Web Access as authentication mechanism.

Table 4 DPI Supported Application Attributes

ID	Application Attribute Name	Description
18	amazon-cloud	Indicates that the current connection is a service of Amazon Cloud.
19	apache	Indicates that the server side is an Apache server.
20	mysql-server	Indicates that the server side is a MySQL database server.
21	mariadb-server	Indicates that the server side is a MariaDB database server.
22	ntlm	Current connection uses NTLM as authentication mechanism.
23	microsoft-windows	Indicates that the client side is the operating system Microsoft Windows.
24	chrome	Indicates that the client side is the operating system Chrome

Protocol/Application Group

Many protocols/applications often provide a similar functionality. For example, both Skype and Viber can be used for VoIP and be classified under Protocol/Application Group “VoiceoverIP”

Table 5 DPI Supported Protocol/Application Group

ID	Group Name	Description
2048	Generic	General group for all not yet assigned protocols and applications
2049	Peer to Peer	Group for Peer 2 Peer related protocols and applications.
2050	Gaming	Group for gaming related protocols and applications.
2051	Tunnel	Group for tunnel related protocols and applications.
2052	Business	Group for business related protocols and applications
2053	E-Commerce	Group for E-Commerce related protocols and applications.
2054	Voice over IP	Group for VoIP related protocols and applications
2055	Messaging	Group for instant messaging related protocols and applications
2056	Streaming	Group for streaming related protocols and applications
2057	Mobile	Group for mobile network related protocols and applications.
2058	Remote Control	Group for remote control related protocols and applications
2059	Mail	Group for mail service related protocols and applications
2060	Network Management	Group for network management related protocols and applications
2061	Database	Group for database related protocols and applications.
2062	Filetransfer	Group for file transfer related protocols and applications.
2063	Web	Group for web surfing related protocols and applications.
2064	Conference	Group for conferencing related protocols and applications.
2065	Social	Group for social media related protocols and applications
2066	Sharehosting	Group for sharehoster related protocols and applications
2067	Deprecated	Group for all deprecated protocols and applications
2068	Industrial	Group for industrial related protocols and applications
2069	Encrypted	Group for encrypted protocols and applications.

Table 5 DPI Supported Protocol/Application Group

ID	Group Name	Description
2070	Advertisement and Analytic Services	Group for protocols and applications which are related to advertisement and analytic services.

Known Anomalies

This section lists the known anomalies in this release

Anomaly	Description
AN-45733	SaferVPN flows are not detected fully.
AN-45607	CounterStrikeGo flows are not detected in MCC.
AN-45606	Briateams application is not detected in MCC when the flow-attribute is Audio.
AN-45605	Briateams application Audio flows are not detected in MCC.
AN-45604	Telegram chat flows are not detected in MCC.
AN-44982	DeviceID based on tcp fingerprinting is returned only for few flows.
AN-40920	DPI: ZenmateVPN not blocked by MCC.
AN-46241	Secure VPN flows are classified as ipsec payload without any app name.
AN-46339	ReceiverRadio flows when browsing the app are not detected.
AN-46521	Lords Mobile flows are detected as unknown.
AN-46704	SOMA application flows are detected without any app name.
AN-47078	HBOGo Help option is detected as Zendesk.
AN-47048	Angry Birds flows are marked as Rovio Gaming.
AN-47050	The Help option in EurosportPlayer applications is undetected.
AN-47099	Textplus application flows are detected as google hangouts for audio call.
AN-47259	DisneyServices (Radio Disney) some of the packets are undetected.
AN-47285	Wechat application audio call application attribute is detected as video.
AN-47451	7TV(Joyn) help option is detected without application name
AN-47442	Watch TV flows are classified as amazon-cloud
AN-47352	GeoCaching(Adventure Lab) major flows are detected as windows azure

Enabling Custom Define Classification (CDC) Support

To enhance release of DPI Signatures, MCC implements the Custom Defined Classification feature with the main DPI Detection Engine (PACE2). PACE2 supports adding custom defined classifications to extend detection without requiring a new PACE2 Detection library. With CDC Signature support, Affirmed achieves reduced the turnaround time for new Application Detection.

Use the following CLI command to enable/disable CDC support:

```
an3000-mcm-slot7cpu1(config)# data-path dpi cdc-classification
[enabled | disabled]
```

By default, this parameter is disabled. Changes to this setting take effect upon next reboot.

In addition, modified the output of the following show command:

```
show ip-data-path dpi dpi-image-version
```

```
ip-data-path dpi dpi-image-version
```

	DETECTION	DETECTION	CDC	DETECTION		
	LIBRARY	LIBRARY	LIBRARY	LIBRARY	DECODER	DPIXML
ID	VERSION	MD5SUM	VERSION	MD5SUM	VERSION	MD5SUM

5	18.11.16	1b024053d0846f841acecaed6de7ff8c	1.0.2			
	e9b6f594b0d9919f506d289c58579a26			7b27466cfaf094a4188f6a7b30026774		